

Spatial transcriptomics technologies

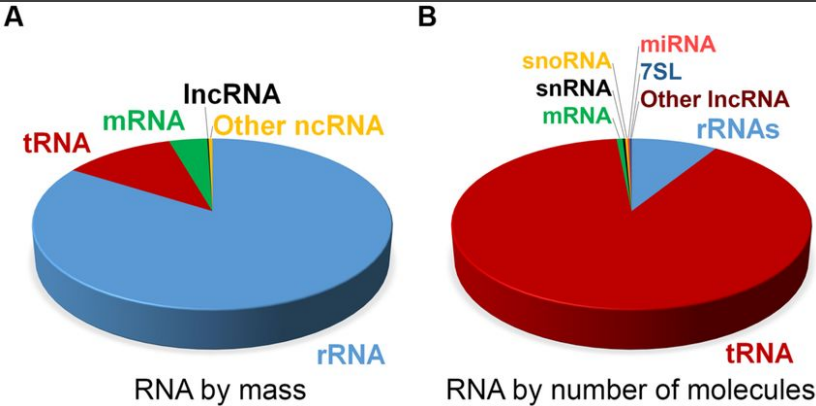
Åsa Björklund

NBIS, SciLifeLab, Chalmers University of Technology

26-10-06

Credit to Lars Born for slides from 2025 course

The cell transcriptome

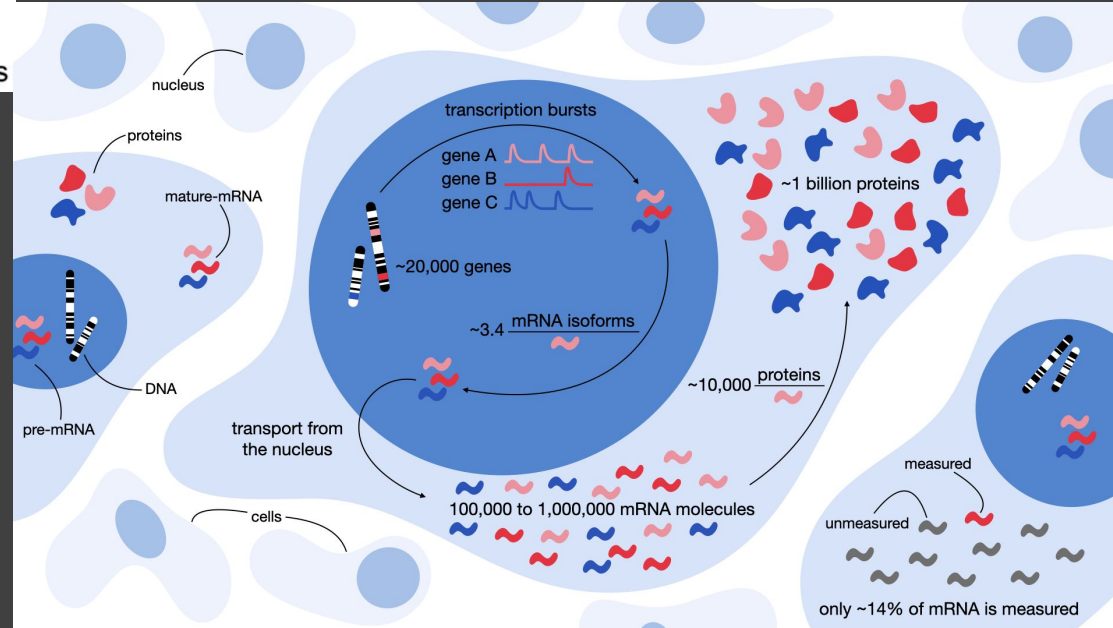


A human cell is on average 10 to 30 μm

- T-cell 7-10 μm
- Fibroblast 10-30 μm
- Neuron 10-100 μm
- Keratinocyte 30-50 μm
- Adipocyte 70-120 μm

Mature mRNAs:

100,000 - 1,000,000 molecules per cell



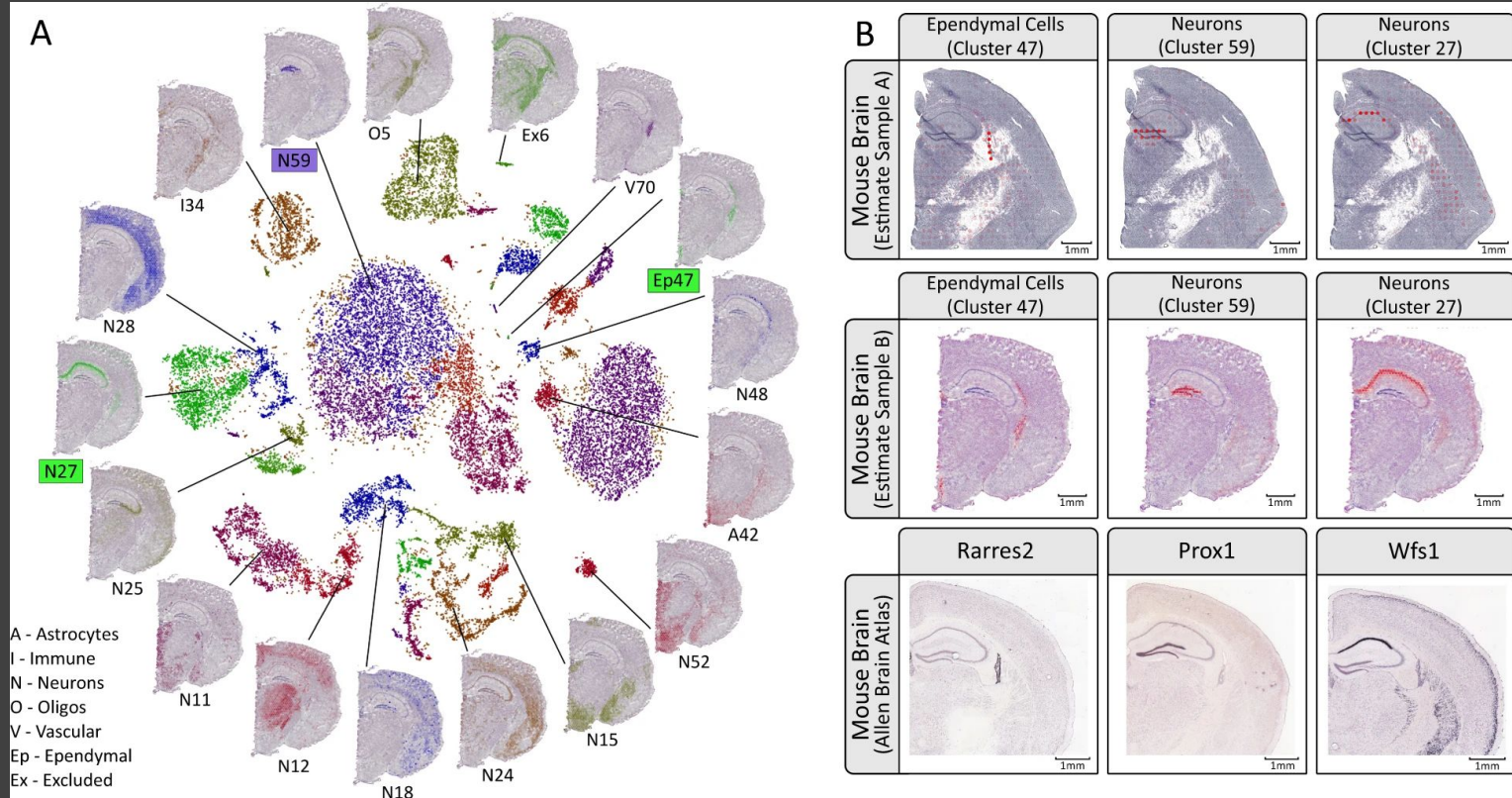
Some terminology

Spatial transcriptomics:

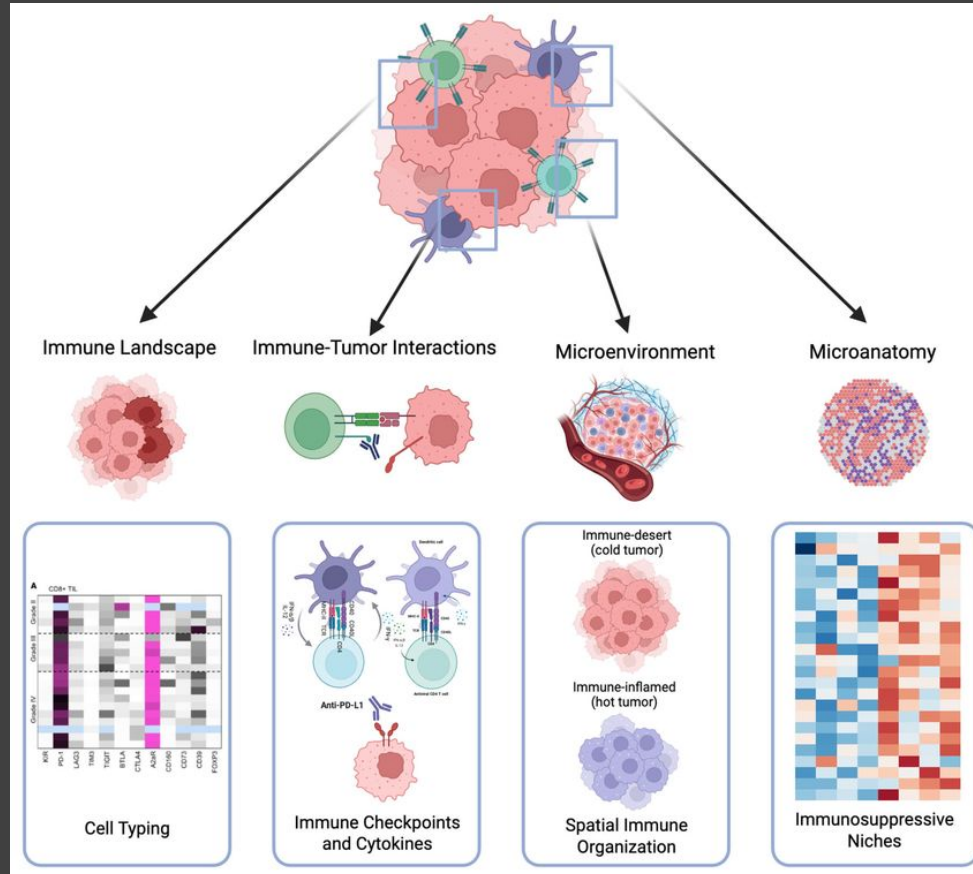
- The study of RNA transcripts in a spatial context.
- A method of sequencing-based spatial transcriptomics

Spatially Resolved Transcriptomics (SRT)

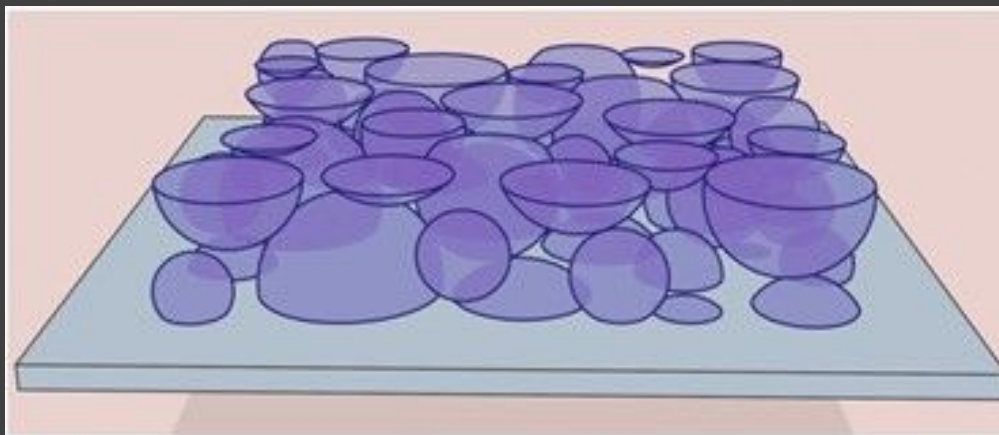
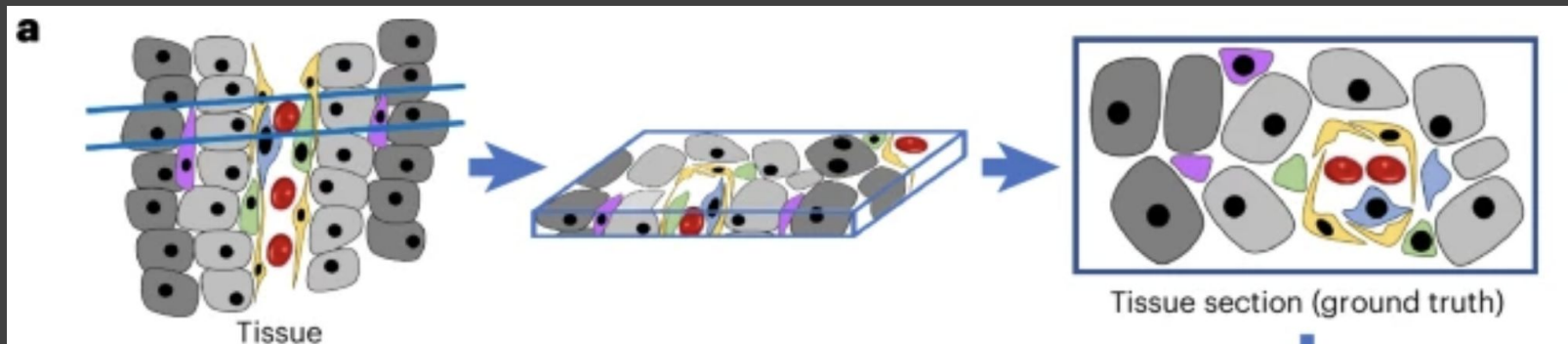
SRT for verification of scRNAseq atlases



SRT to study cells in a tissue context

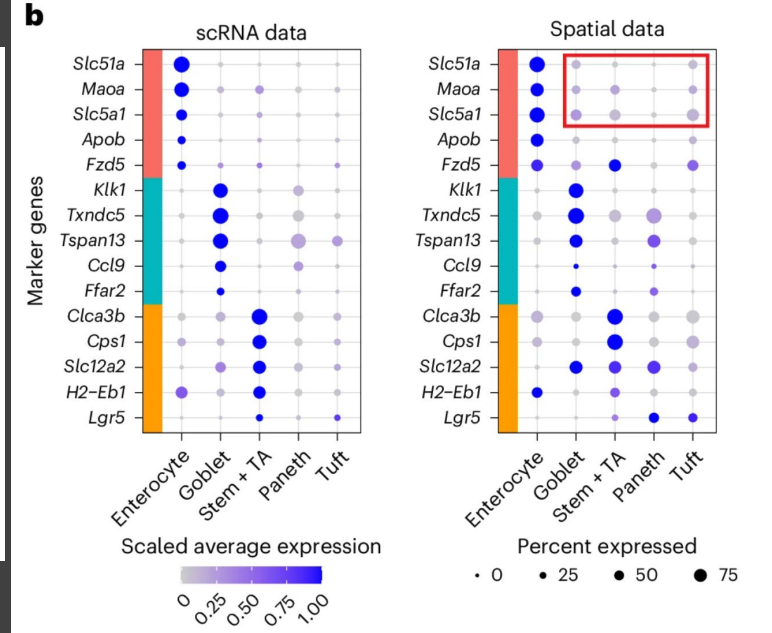
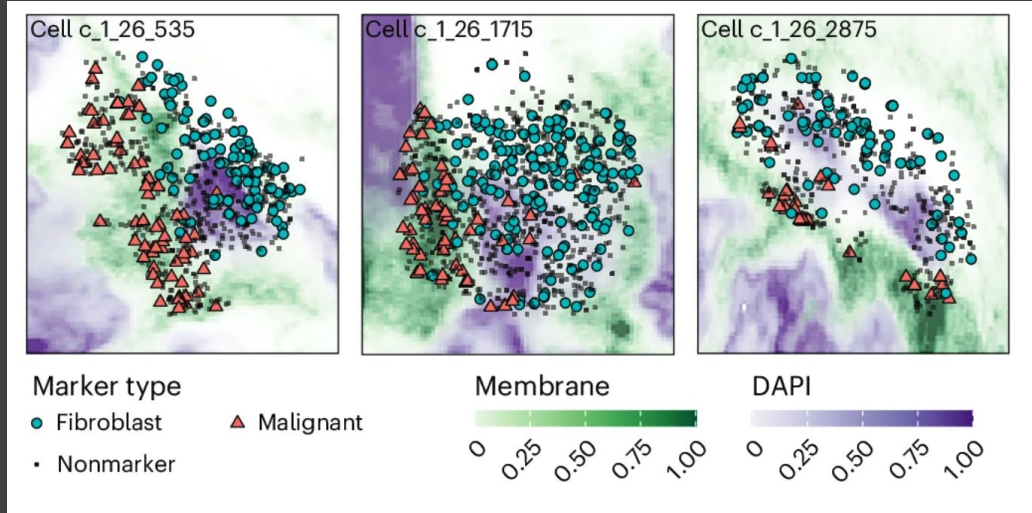


SRT is NOT the same as scRNAseq in a spatial context



(Si et al. Nature Methods 2024, <https://ovrlpy.readthedocs.io/>)

The cell segmentation challenge



(Mitchel et al. Nature Genetics 2026)

Goal

Understanding spatial technologies:

- Choose the right technology
- Know the limitations / biases
- Recognize technical artifacts

Overview of main approaches



Resolution



Sensitivity



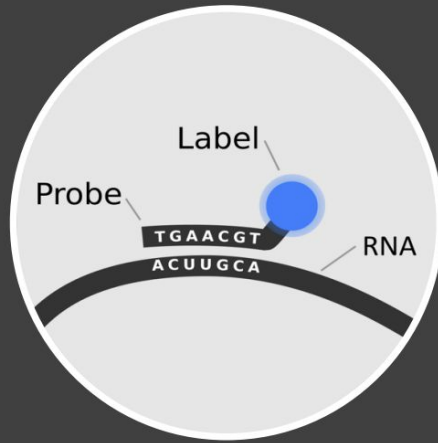
Number of targets



Area

Spatial RNA detection

Microscopy

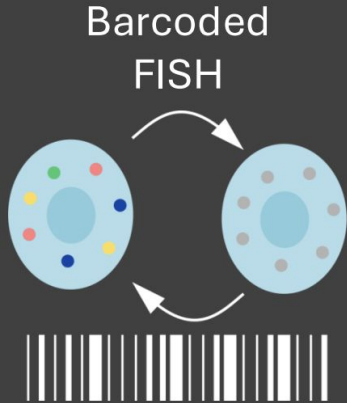


in situ Hybridization

Sequencing

Spatial RNA detection

Microscopy



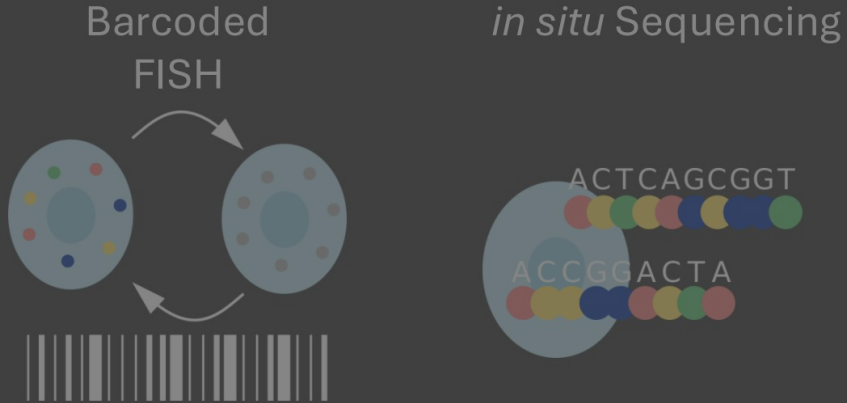
in situ Sequencing



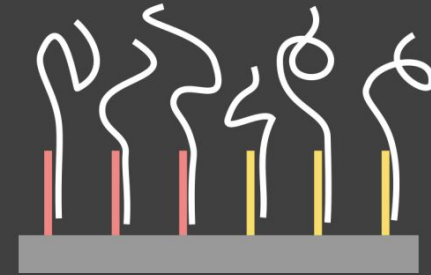
Sequencing

Spatial RNA detection

Microscopy



Sequencing



Spatial barcodes

Spatial RNA detection

Microscopy

Barcoded
FISH

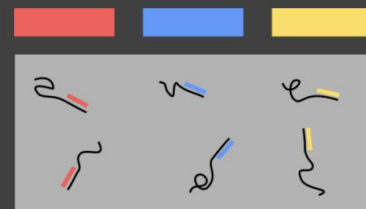
in situ Sequencing



Sequencing

Spatial Sequencing

Spatial tagging

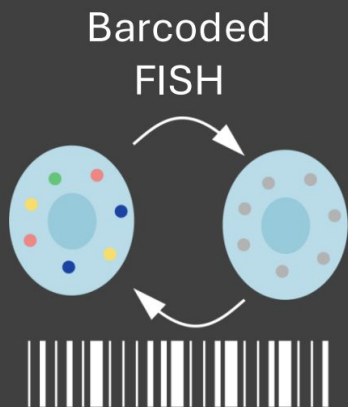


RNA moves

Barcodes move

Spatial RNA detection

Microscopy



Targeted

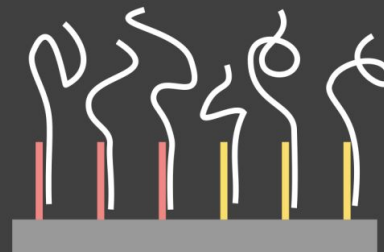
in situ Sequencing



Targeted &
Un-targeted

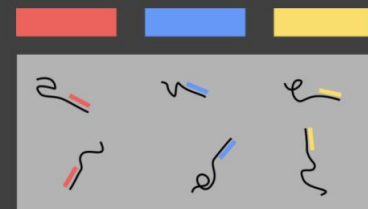
Sequencing

Spatial Sequencing



Targeted &
Un-targeted

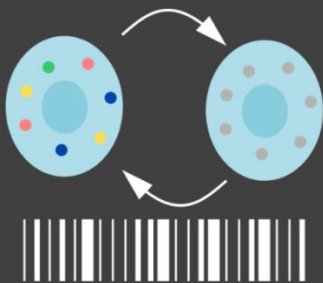
Spatial tagging



Un-targeted

Spatial RNA detection

Barcoded FISH



- MERFISH
- Vizgen - MERFISH
- seqFISH
- Spatial Genomics - GenePS
- EEL-FISH
- HybISS
- 10X - Xenium
- Nanostring/Bruker - CosMx
- Resolve - Mol. Cartography

in situ Sequencing



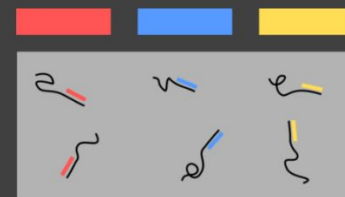
- ISS
- STARmap
- StellarOmics
- Singular genomics - G4X

Spatial Sequencing



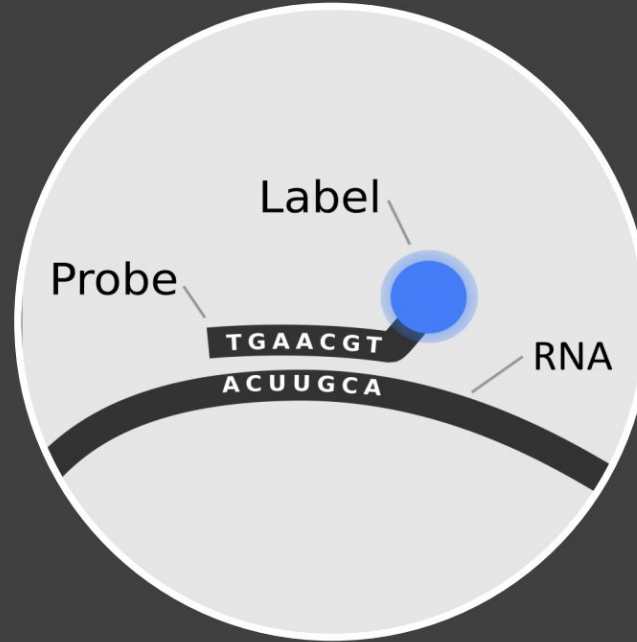
- Spatial Transcriptomics
- 10X - Visium (HD)
- Slide-seq
- Curio - Seeker
- Stereo-seq
- BGI STOmics - Stereo-seq
- Seq-Scope, Open-ST, Nova-ST

Spatial tagging



- DBiT-seq
- AtlasXomics
- Slide-tags
- Curio - Trekker

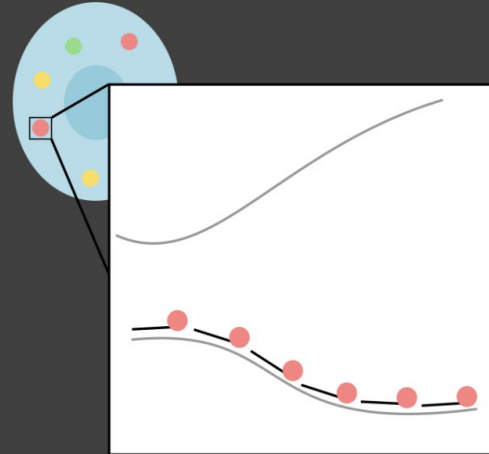
Fluorescent in situ Hybridization (FISH)



single molecule FISH (smFISH)

Probes

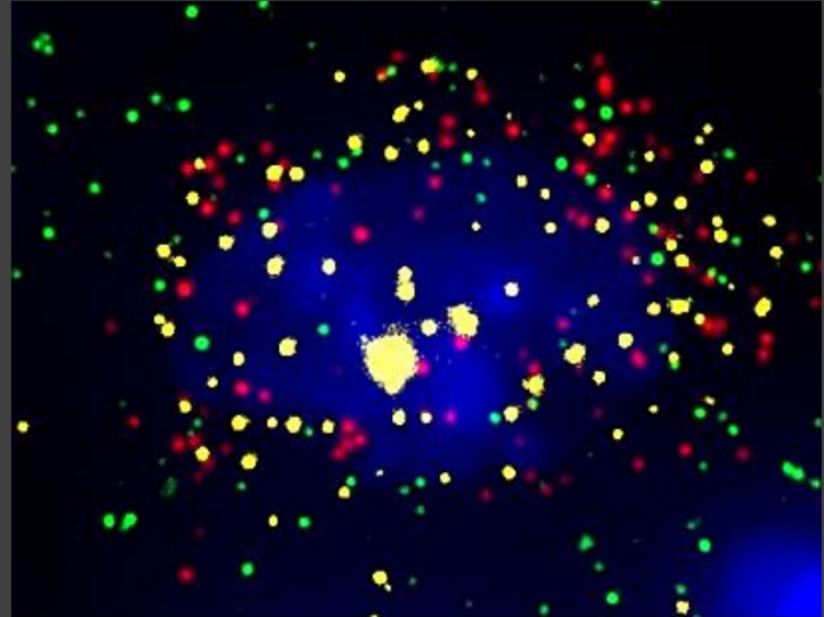
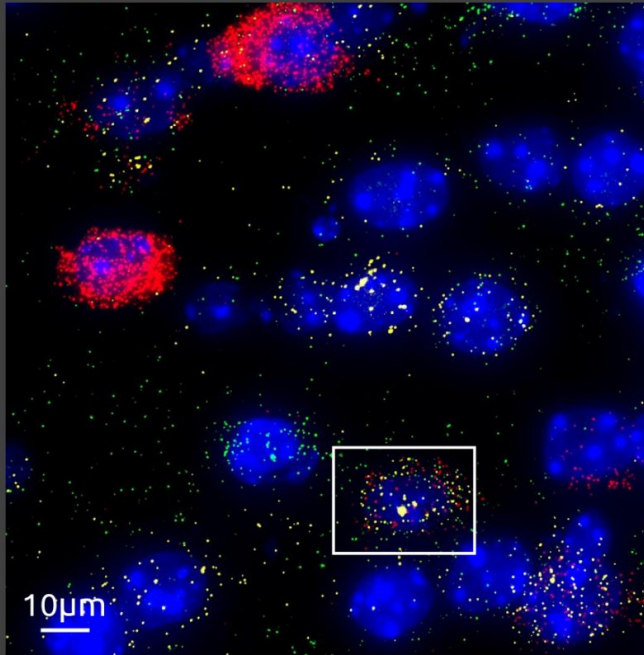
Gene 1	40X	
Gene 2	40X	
Gene 3	40X	
Gene 4	40X	



Femino *et al.* Science 1998
Raj *et al.* Nature Methods 2008

single molecule FISH (smFISH)

Gene 1 Gene 2 Gene 3 DNA



Limited by number of flourophores

~20.000 genes

4-7 colors

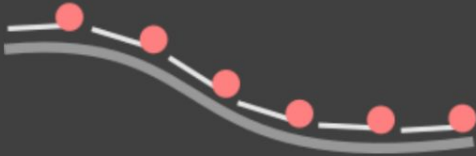
Solution:

- Repeated staining on same sample
- Barcoding



Reprobing same molecule

Round 1



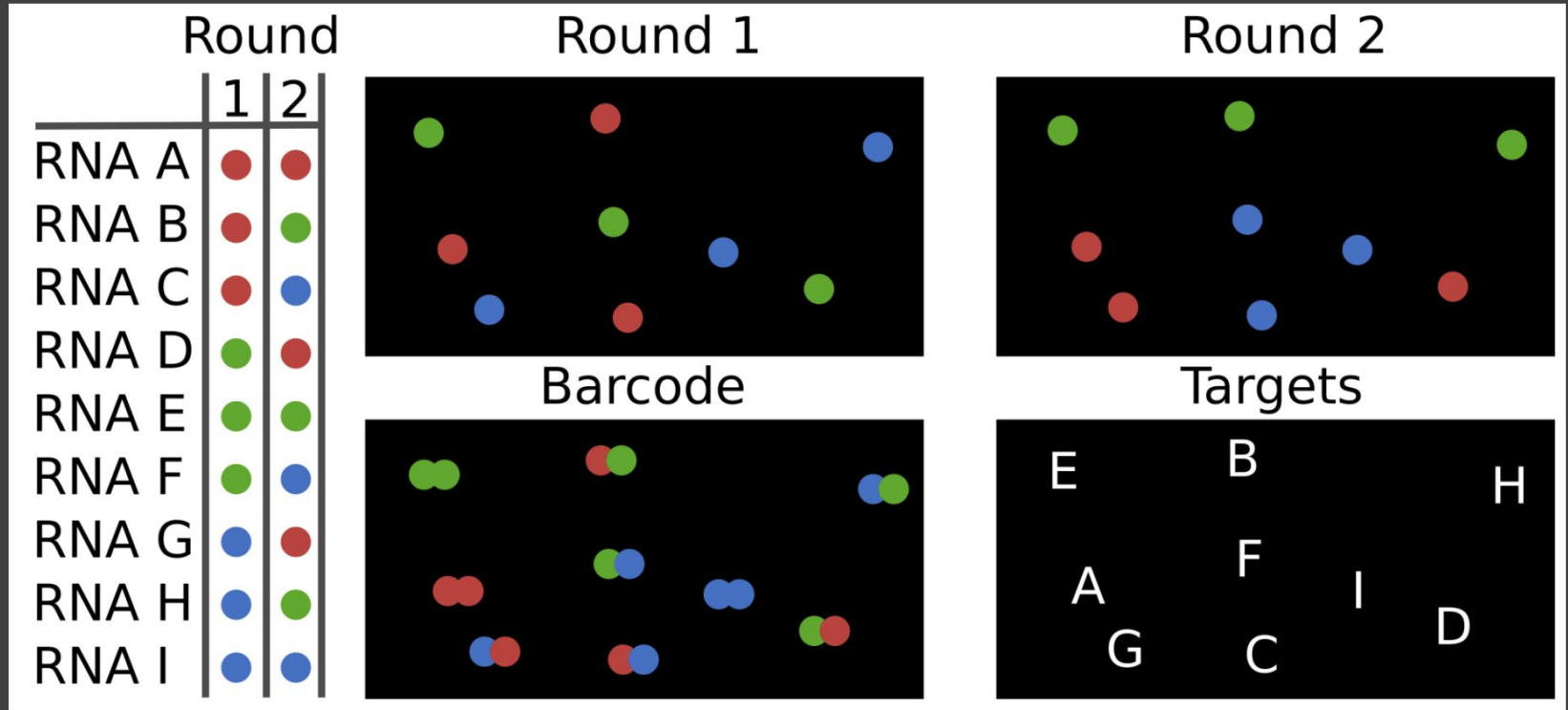
Stripping



Round 2



Barcoding



Lubeck et al. Nature Methods 2014

Barcoding

Scaling:

Number of rounds n

Number of flourophores f

$$\text{Targets} = f^n$$

$$4^8 = 65,536$$

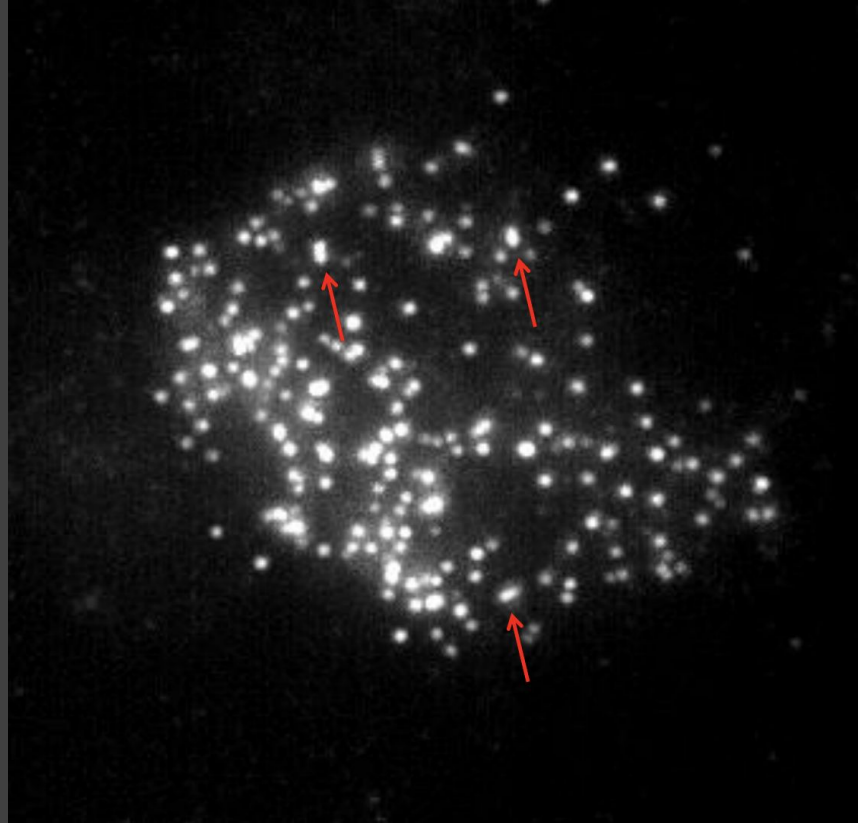
Barcoding

Problems:

- Optical density
- Errors

Solution:

- Sparse barcodes



Barcoding

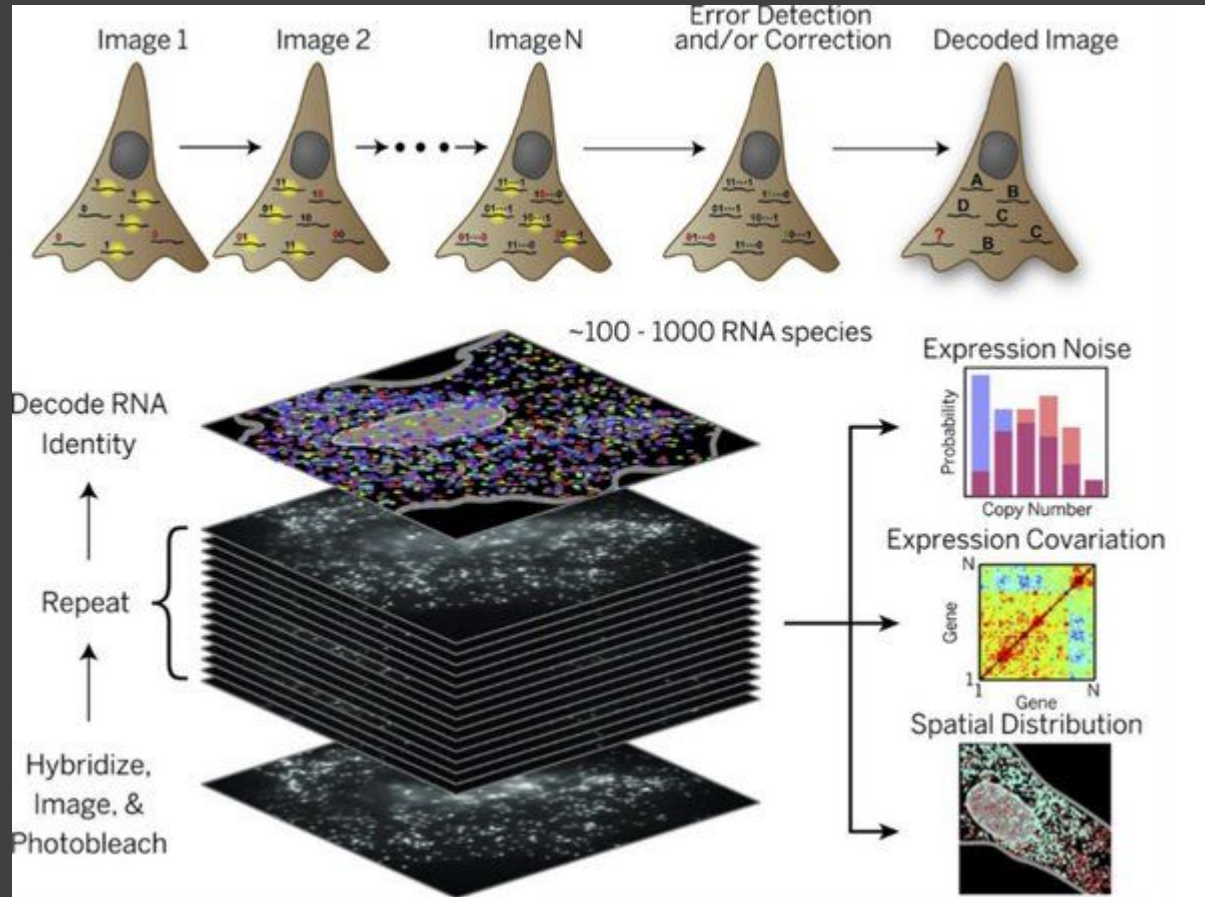
Sparse barcodes

100110001

Error robustness

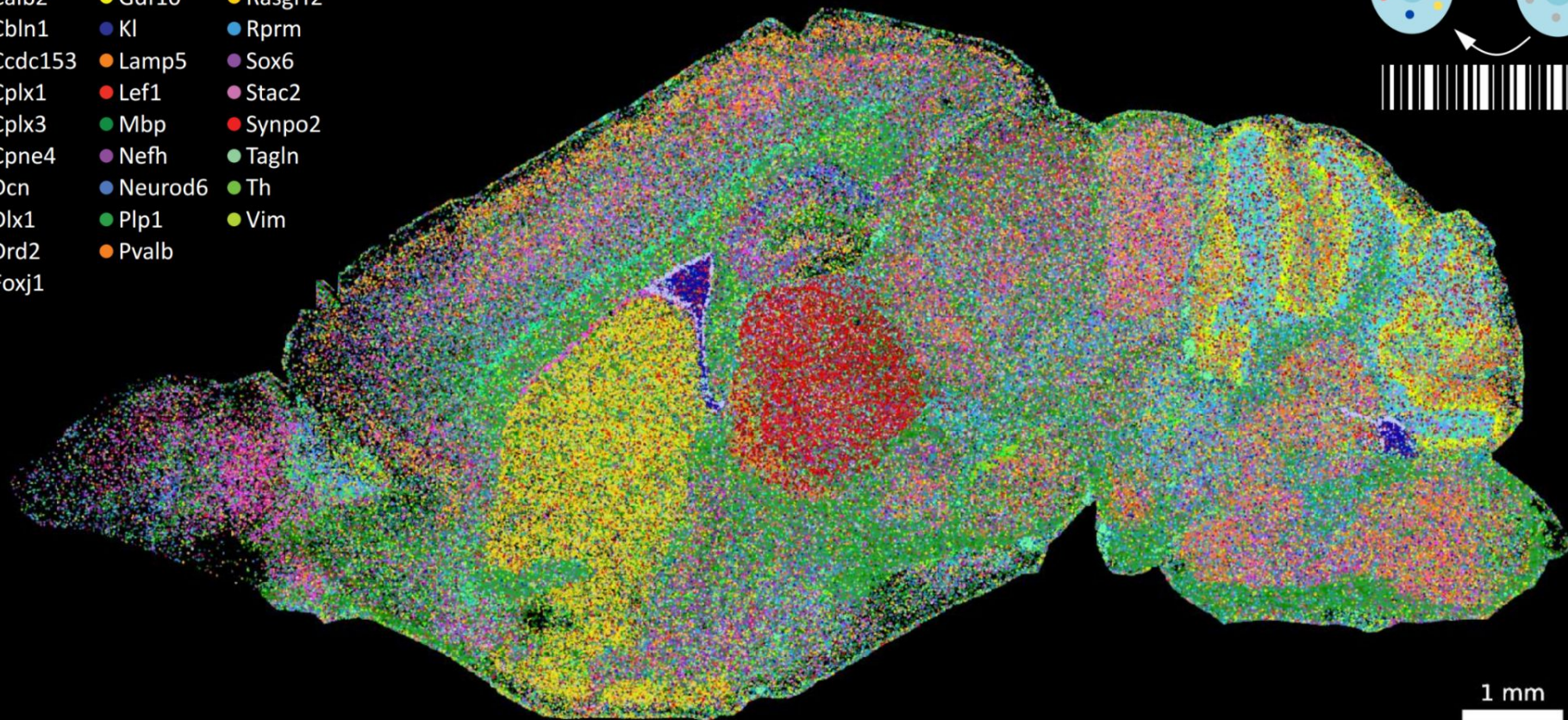
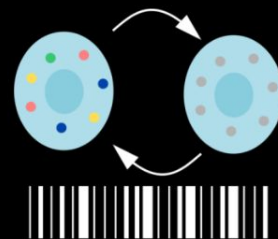
MERFISH

Chen et al. 2015 Science



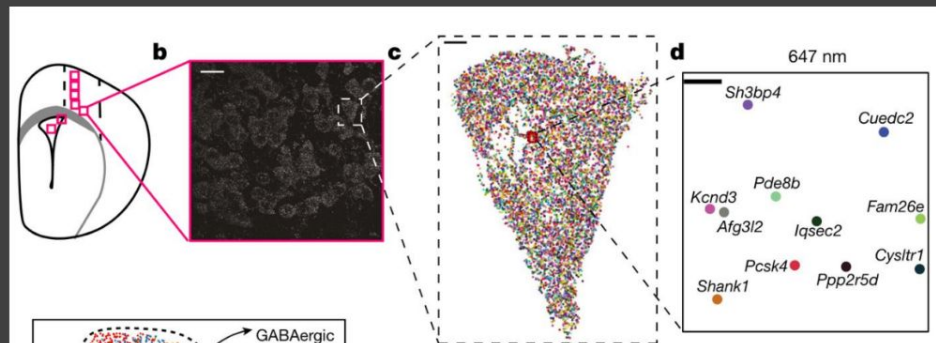
30 Selected genes

● Adora2a	● Gabra6	● Ramp3
● Calb2	● Gdf10	● Rasgrf2
● Cbln1	● Kl	● Rprm
● Ccdc153	● Lamp5	● Sox6
● Cplx1	● Lef1	● Stac2
● Cplx3	● Mbp	● Synpo2
● Cpne4	● Nefh	● Tagln
● Dcn	● Neurod6	● Th
● Dlx1	● Plp1	● Vim
● Drd2	● Pvalb	
● Foxj1		



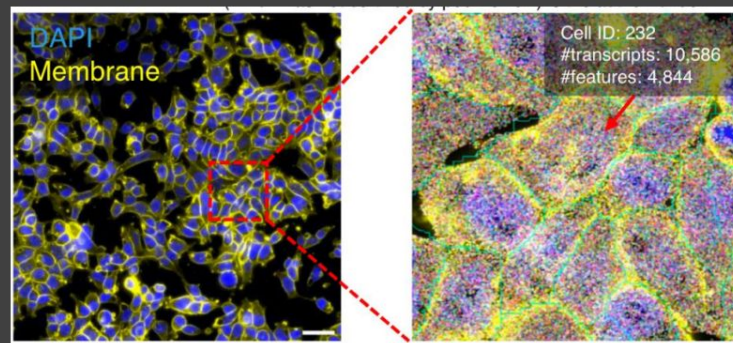
1 mm

seqFISH+



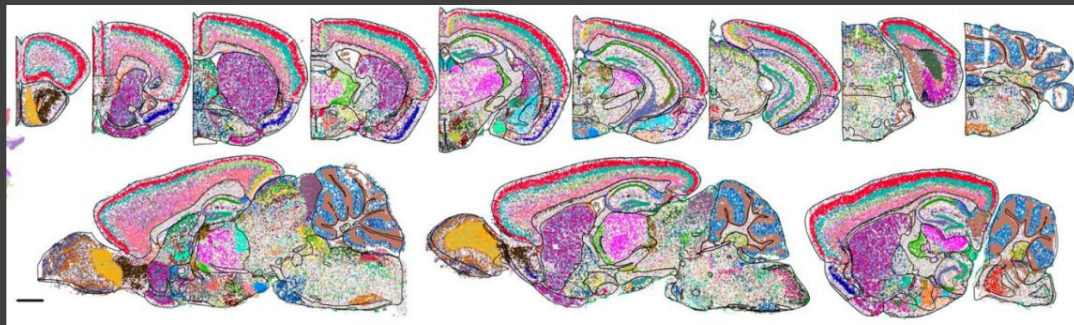
Eng *et al.* 2019 Nature. 10,000 plex

CosMx



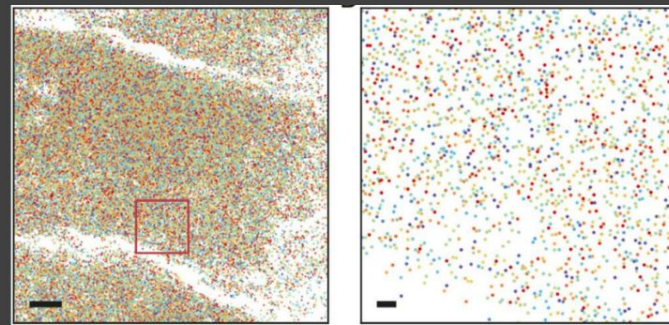
Khafizov *et al.* 2024 BioRxiv. 18,993 plex

MERFISH



Zhang *et al.* 2023 Nature. Full mouse brain

MERFISH



Xia *et al.* 2019 PNAS. 10,000 plex

Barcoded smFISH



Methods: MERFISH, seqFISH, EEL FISH

Companies: Vizgen, Spatial Genomics, Nanostring/Bruker, Resolve



Resolution: Diffraction limited (150-300nm)



Detection efficiency: 70-90% *

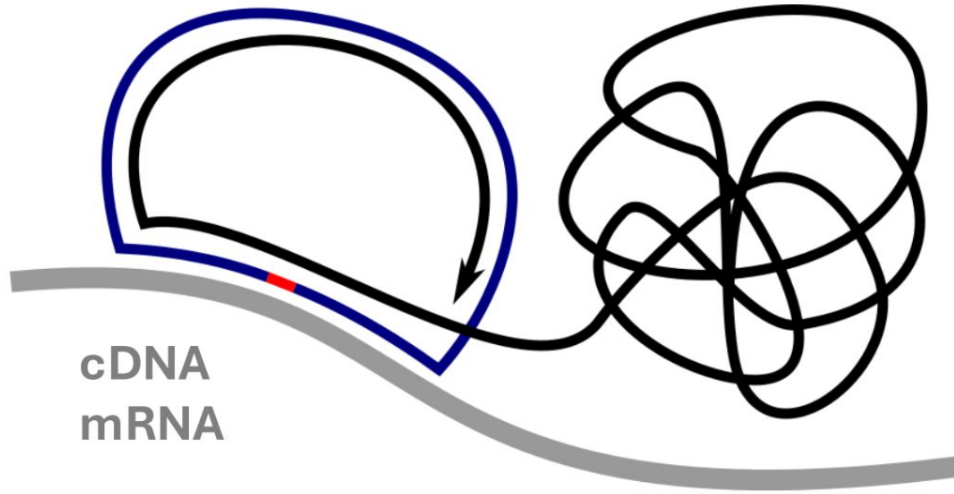


Gene throughput: 100 - 19,000

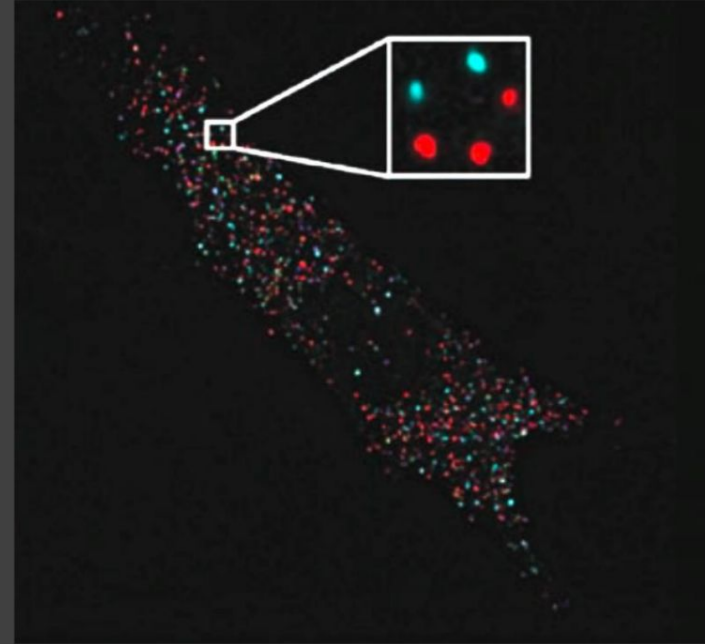


Spatial throughput: several mm² - cm²

Rolling circle amplification (RCA)

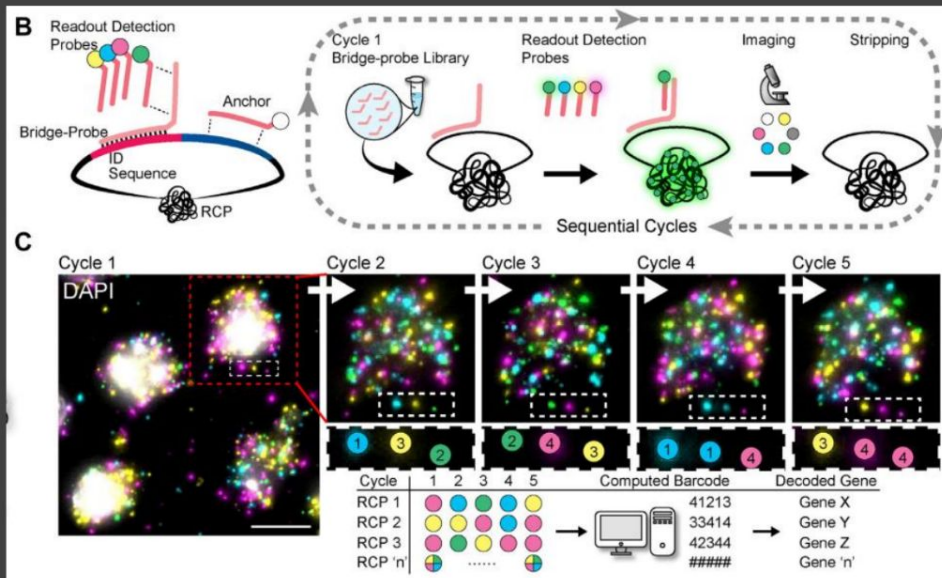


Padlock probe, Targeted, Enzymatic amplification



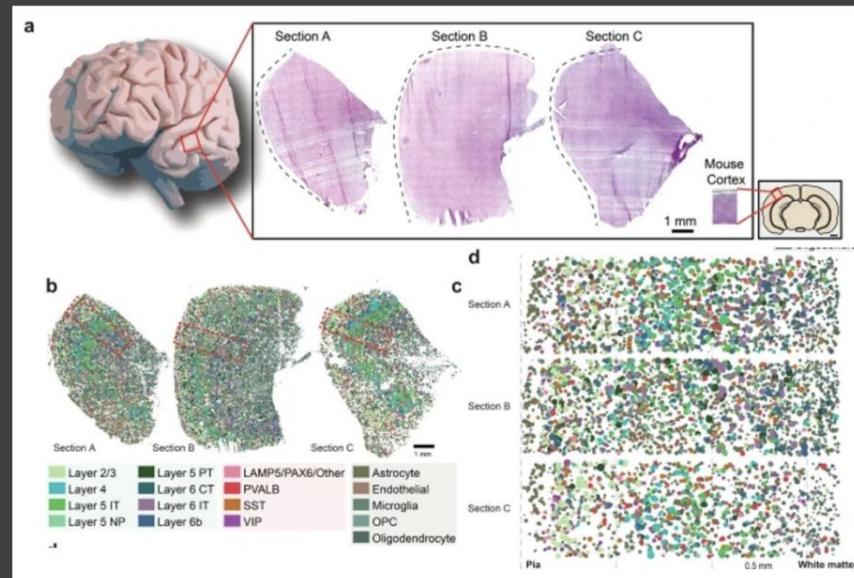
RCA amplified barcoded FISH

HybISS



Gylborg *et al.* 2020 Nucleic Acids Research

HybISS



Mattsson Langseth *et al.* 2021 Nature Communications Biology

Amplified barcoded smFISH



Methods: HybISS, HybRISS

Companies: 10X Xenium



Resolution: Amplicon size $\sim 0.5\text{-}1\text{ }\mu\text{m}$



Detection efficiency: 10 - $\sim 50\%$



Gene throughput: 100 – 1,000

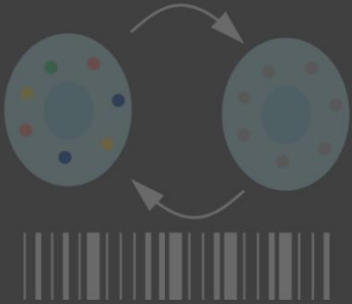


Spatial throughput: cm^2 - several cm^2

spatial RNA detection

Microscopy

Barcoded
FISH



in situ Sequencing



Sequencing

Spatial Sequencing

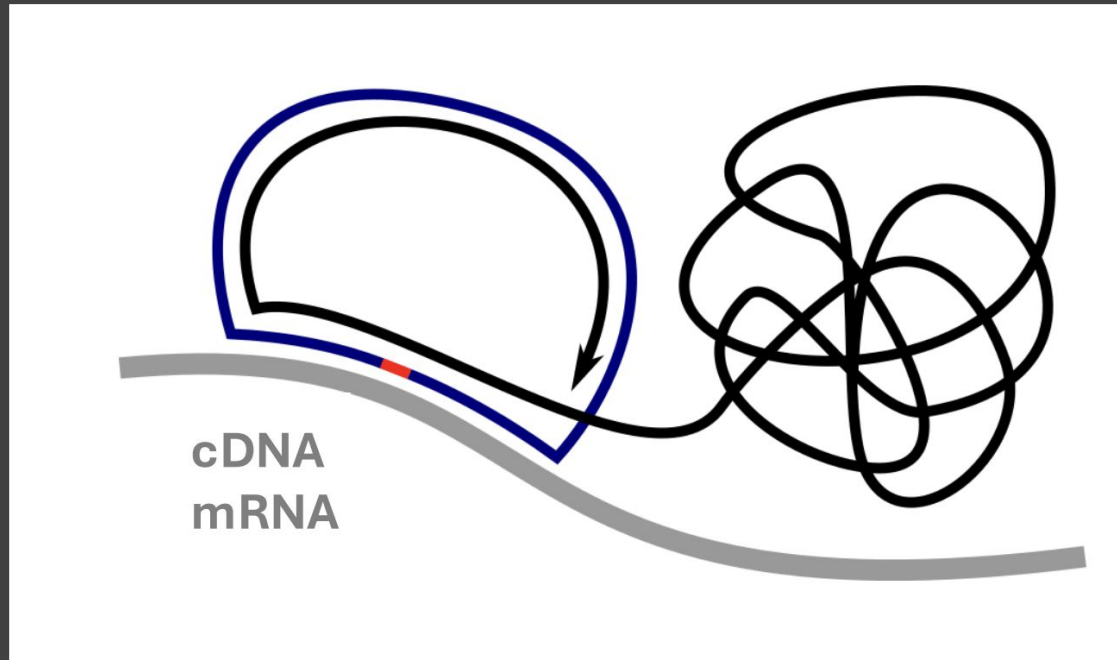


Spatial tagging



in situ sequencing (ISS)

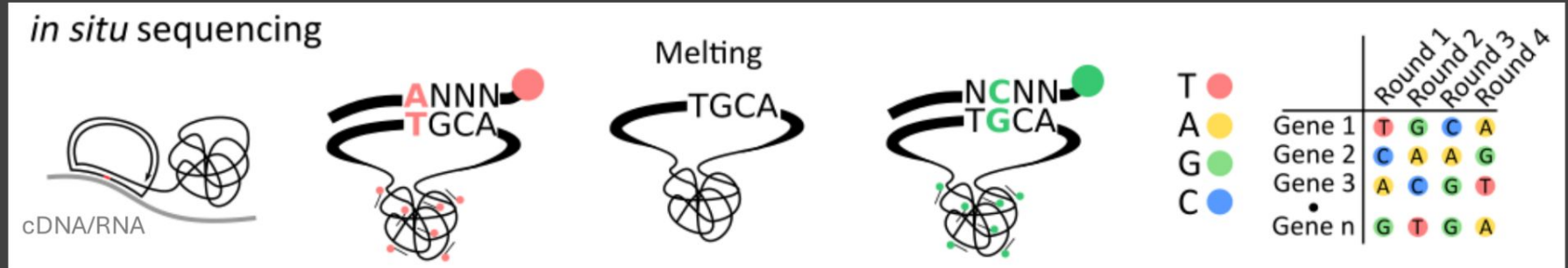
Rolling circle amplification



cDNA
mRNA

Padlock probe. Targeted

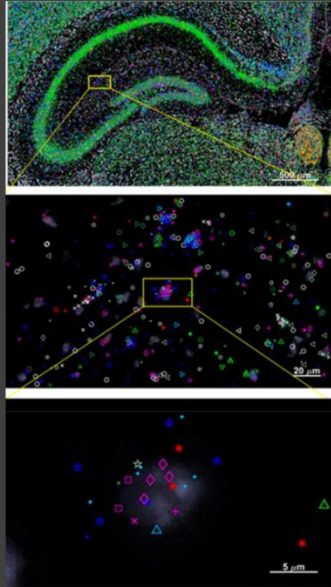
in situ sequencing (ISS)



Sequencing by ligation

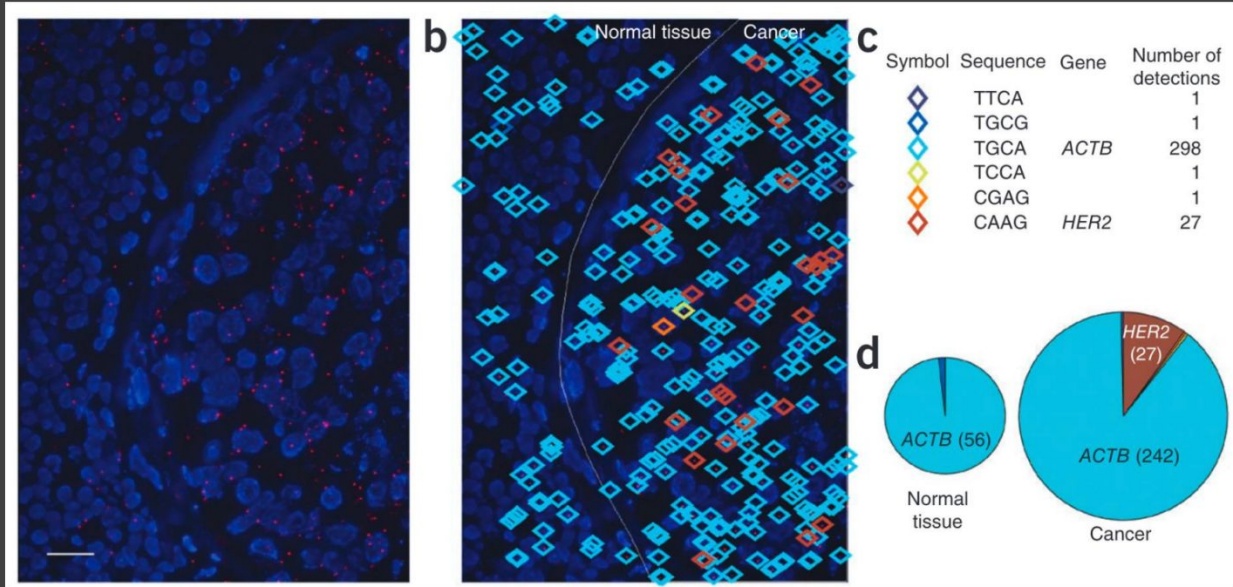
in situ sequencing (ISS)

Barcode sequencing



Qian *et al.* 2020 Nature Methods

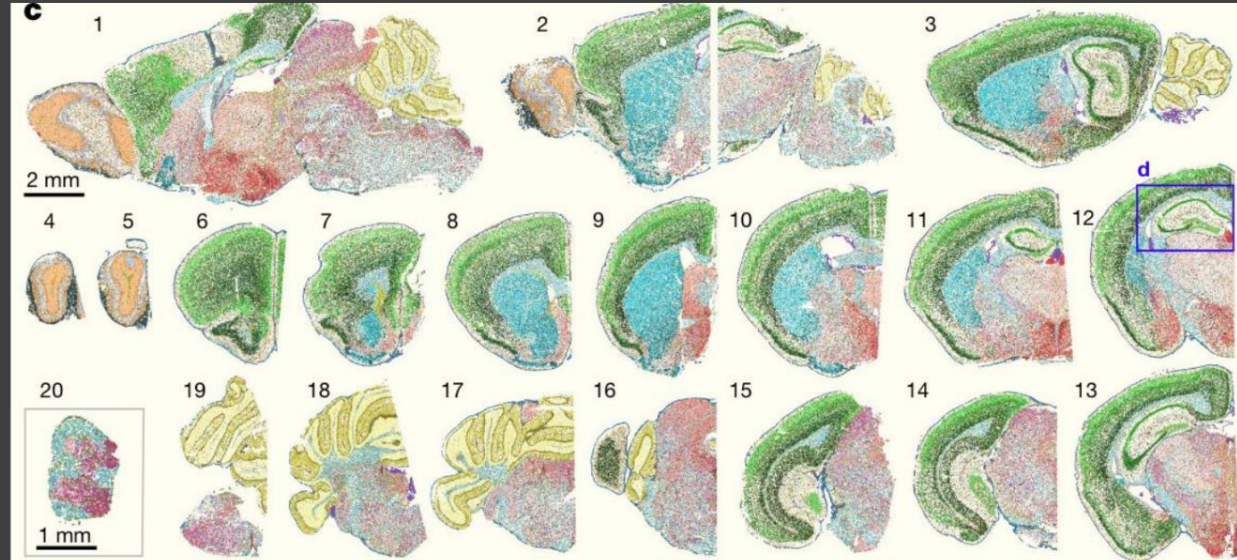
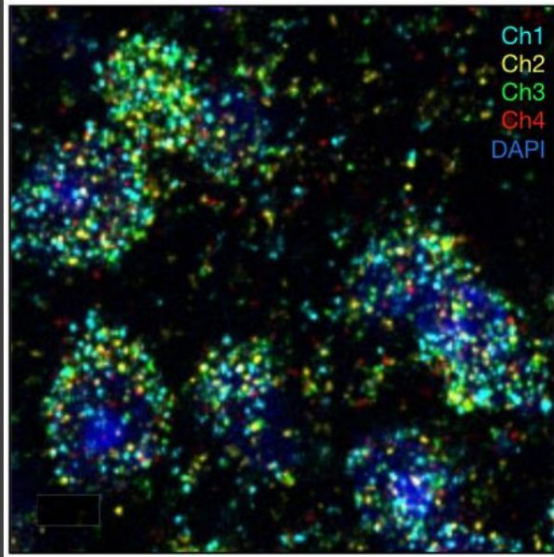
De-novo sequencing



Ke *et al.* 2013 Nature Methods

ISS is the predecessor of Xenium

STARmap



Hailing *et al.* 2023 Nature

SEDAL = Sequencing with Error-reduction by Dynamic Annealing and Ligation
Two-base encoding (similar to SOLiD sequencing)

Sequencing *in situ*



Methods: ISS, STARmap.

Commercial: (Xenium), StellarOmics



Resolution: Amplicon size (0.5 - 1 μm)



Detection efficiency: 10 - ~50%



Gene throughput: 10 - 1,000



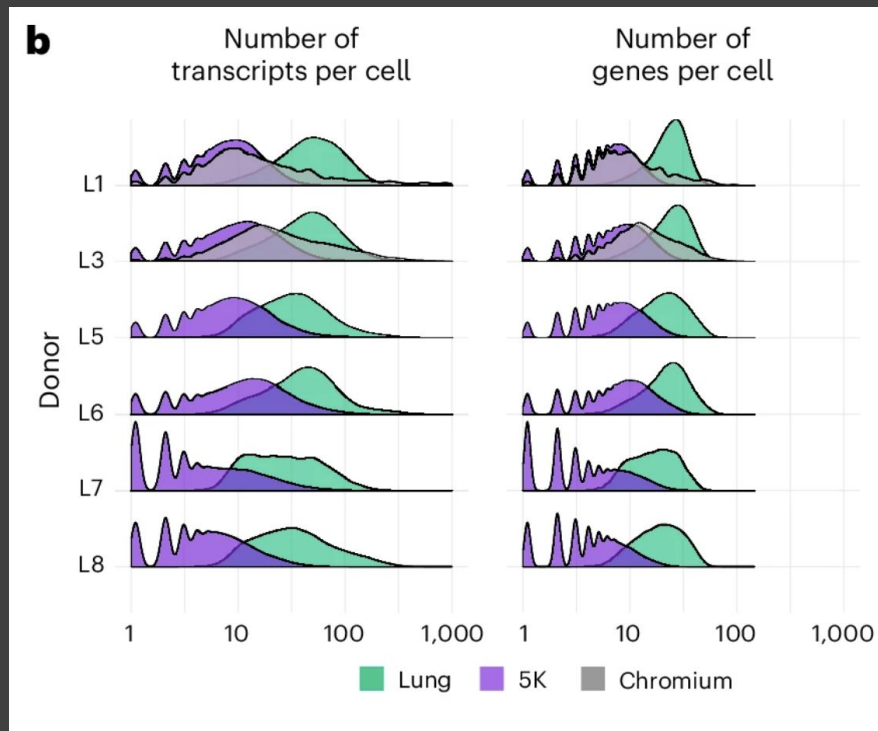
Spatial throughput: several mm^2 - several cm^2

Xenium targeted panels vs Prime 5K panel

Much higher detection with the targeted panel.

Fewer probes per gene to avoid crowding.

Comparison of number of transcripts per cell for 194 common genes between Lung panel, Xenium 5K panel and 10x Chromium snRNAseq.



Bilous et al. Nature Methods (2026)

10X Atera instead of Xenium?

18K genes (WTA) or panels with 1000 genes.

Released in May 2026

Atera In Situ

NOW SHIPPING

**High throughput.
Whole transcriptome.
Single cell sensitivity.**

Never choose between spatial resolution and gene plex again.

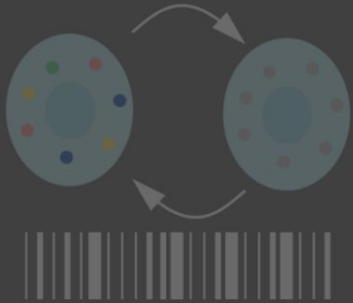
[Speak to an Atera sales executive](#)



spatial RNA detection

Microscopy

Barcoded
FISH

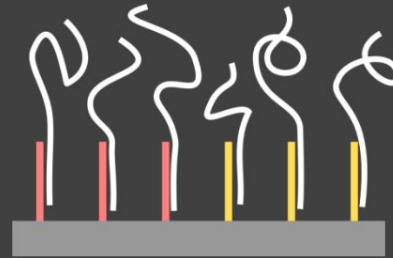


in situ Sequencing



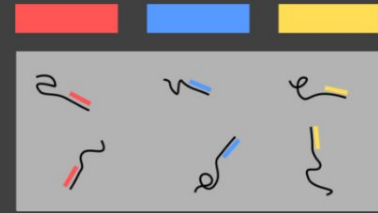
Sequencing

Spatial Sequencing



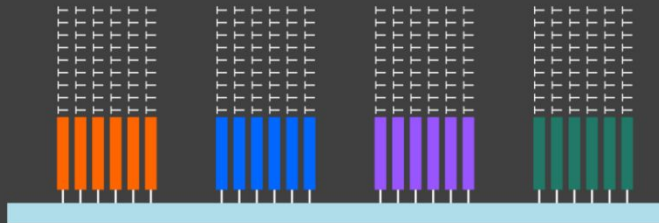
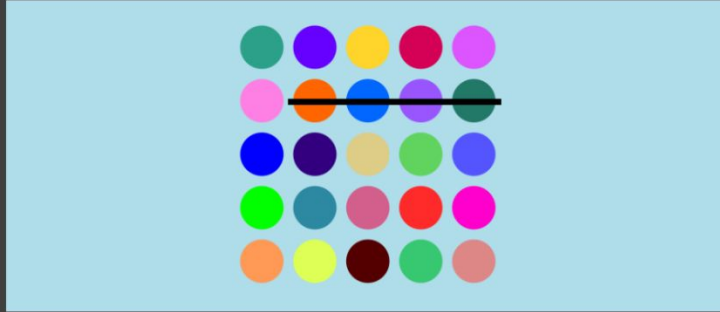
RNA moves

Spatial tagging

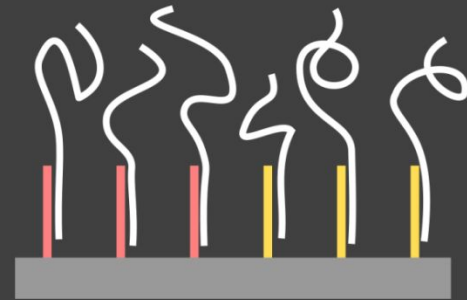


Barcodes move

Spatial Transcriptomics

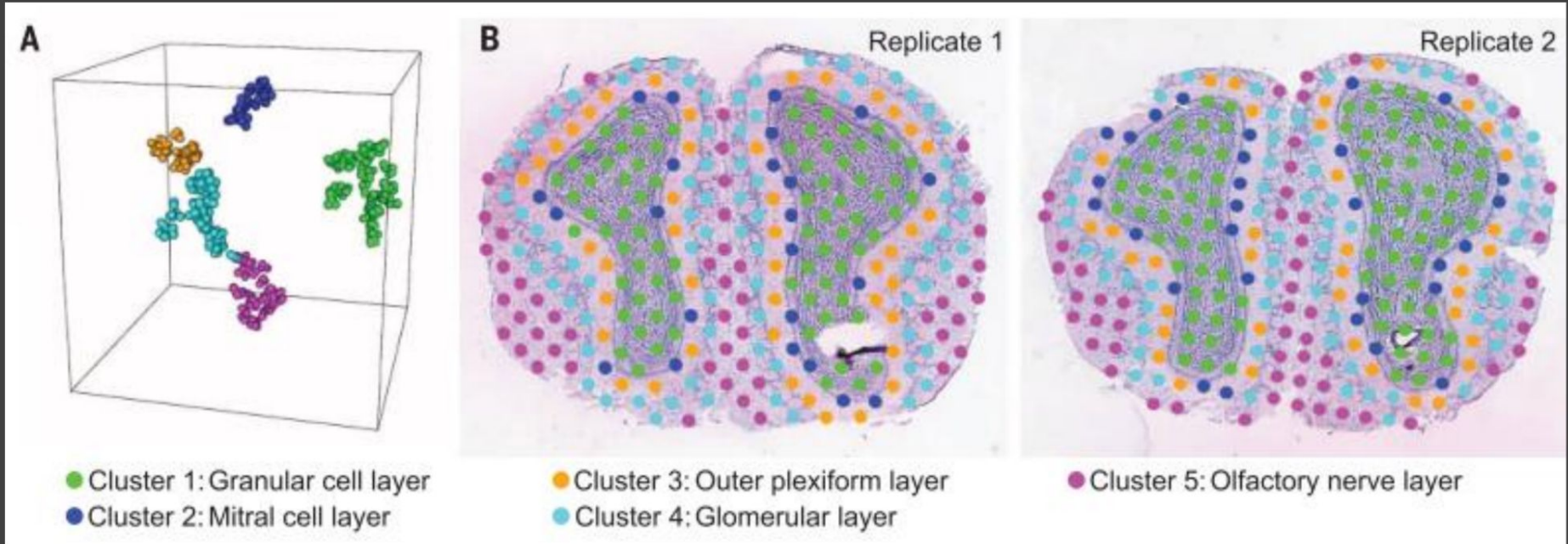


Microarray with spatial barcodes



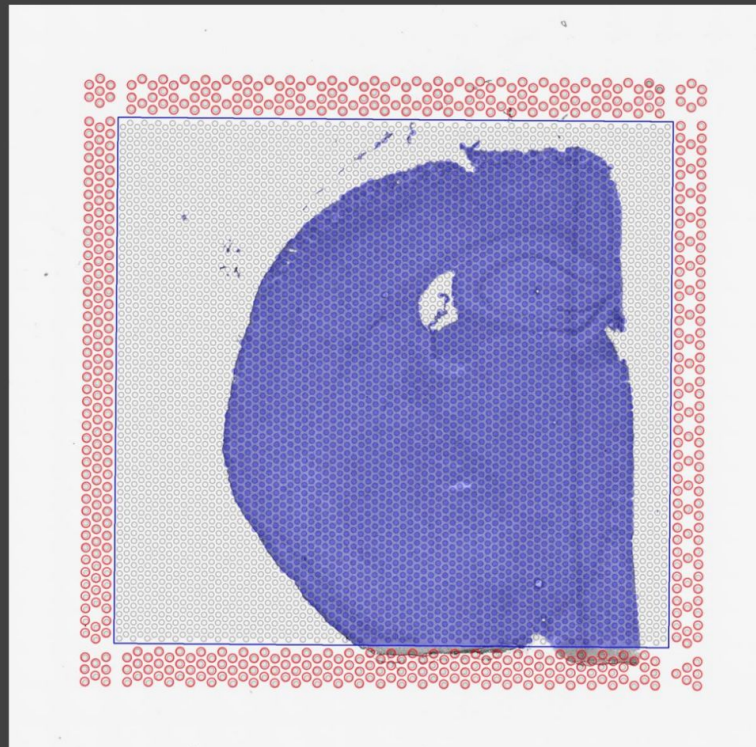
Stahl *et al.* Science 2016

Spatial Transcriptomics

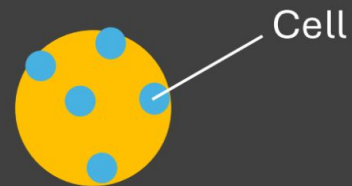


Stahl *et al.* Science 2016

10x Visium



10X Visium

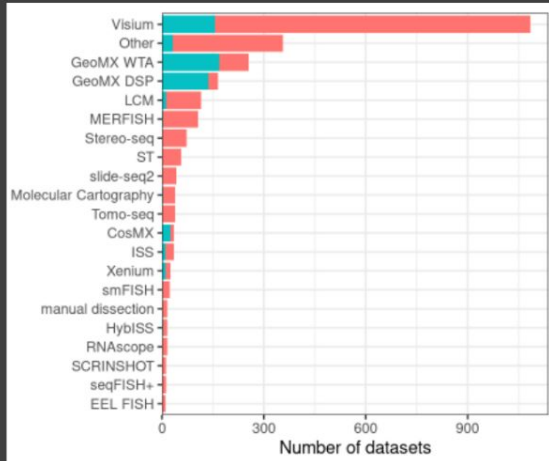


Spot size: 55 μm

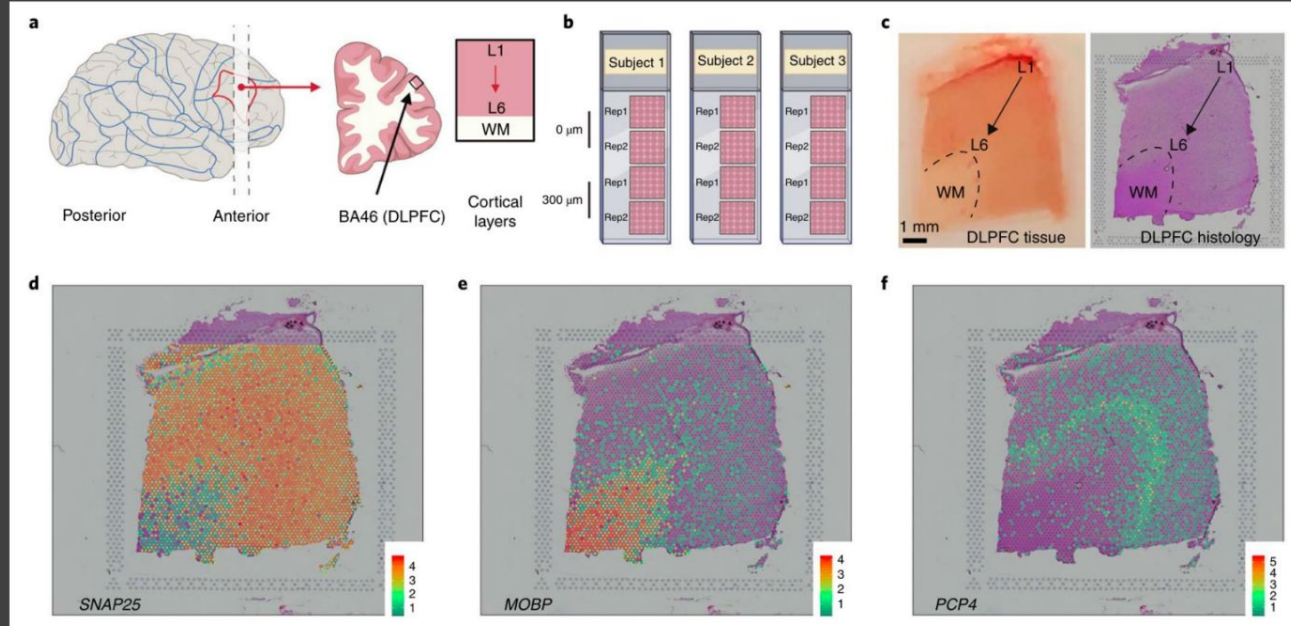


Spot spacing: 100 μm

10x Visium

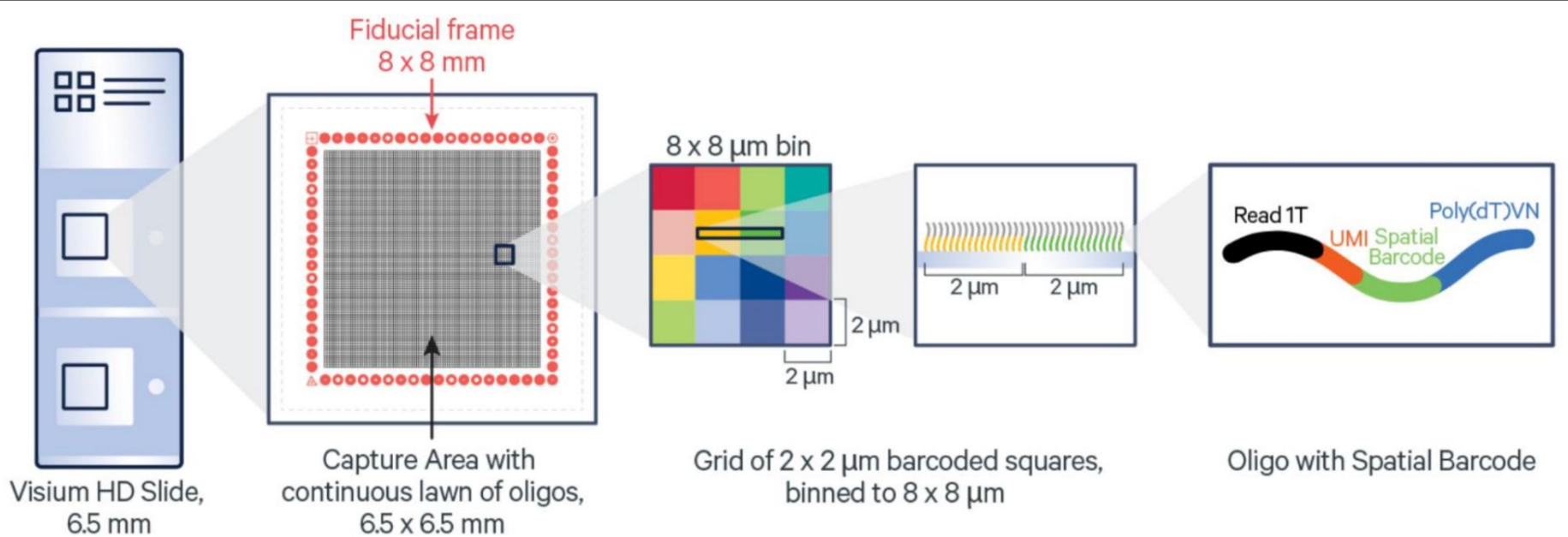


pachterlab.github.io/LP_2021/



Maynard et al. 2021 Nature Neuroscience

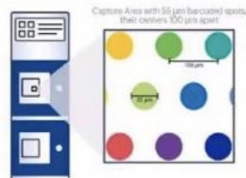
10x VisiumHD



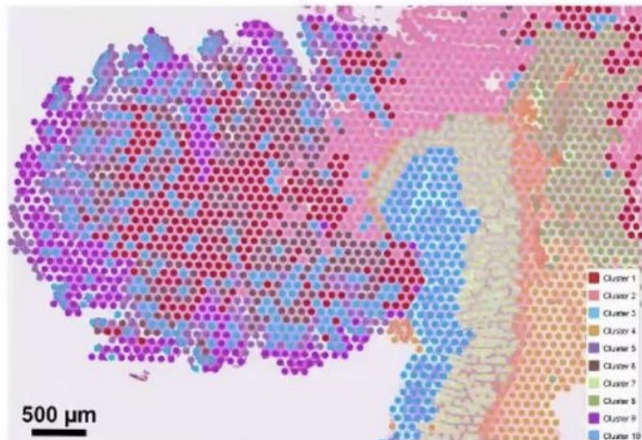
10X Visium HD

10x VisiumHD

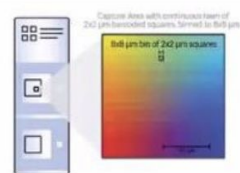
Visium v2



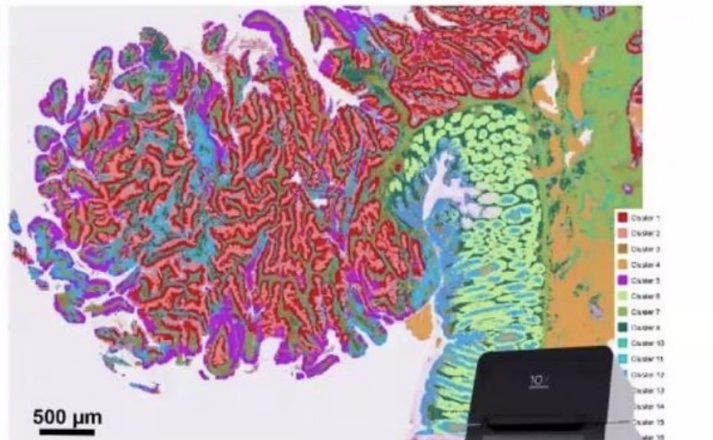
5000 spots
55µm resolution
spaced 100µm



Visium HD

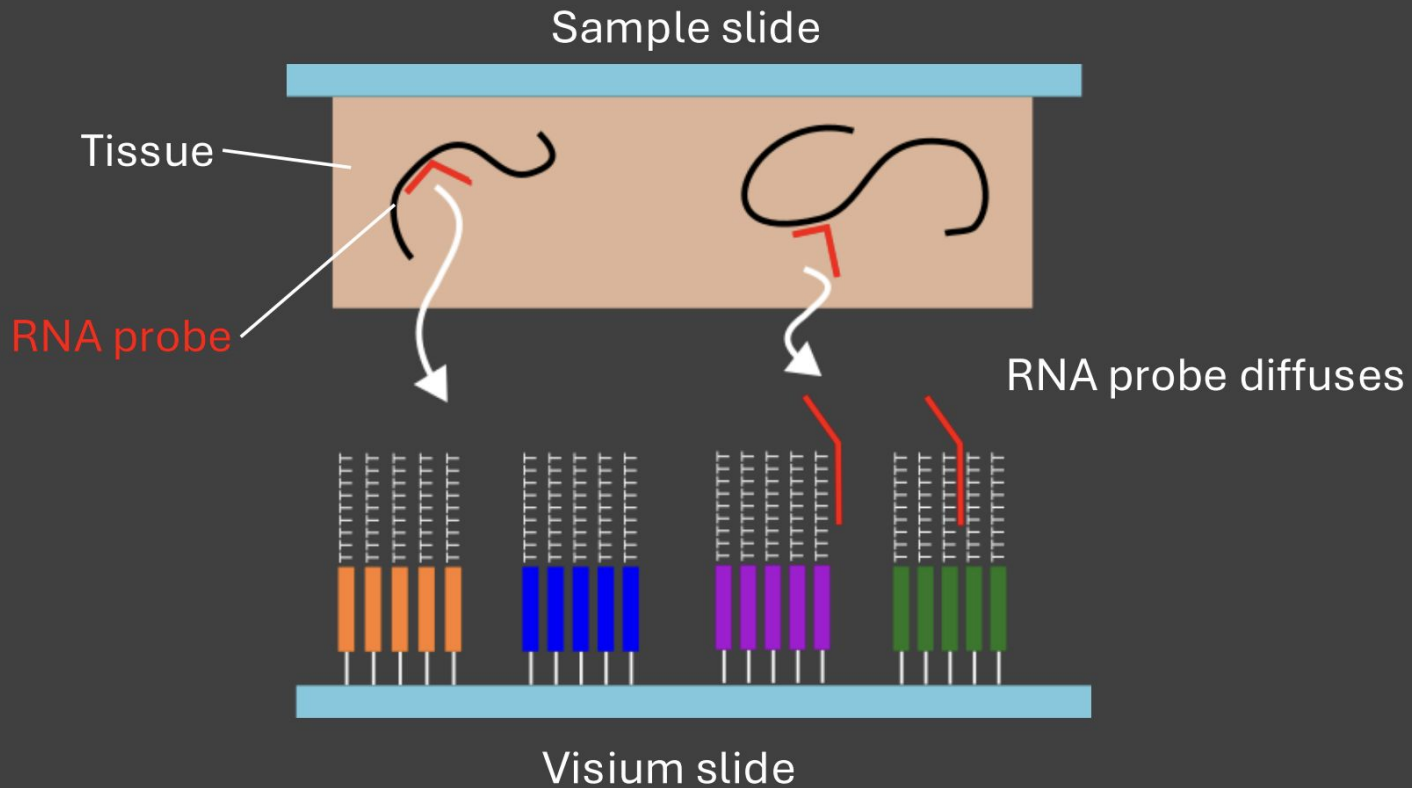


11M squares
2x2 µm resolution
Continuous coverage

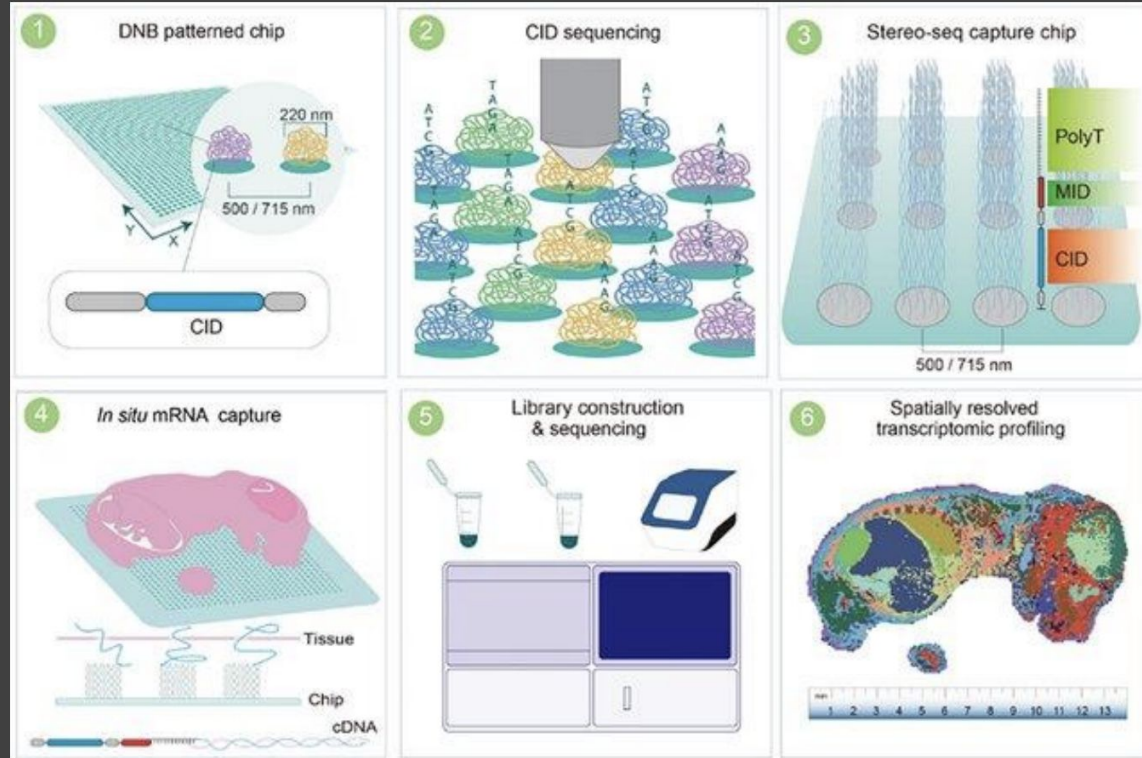


10X Visium HD

10X CytAssist



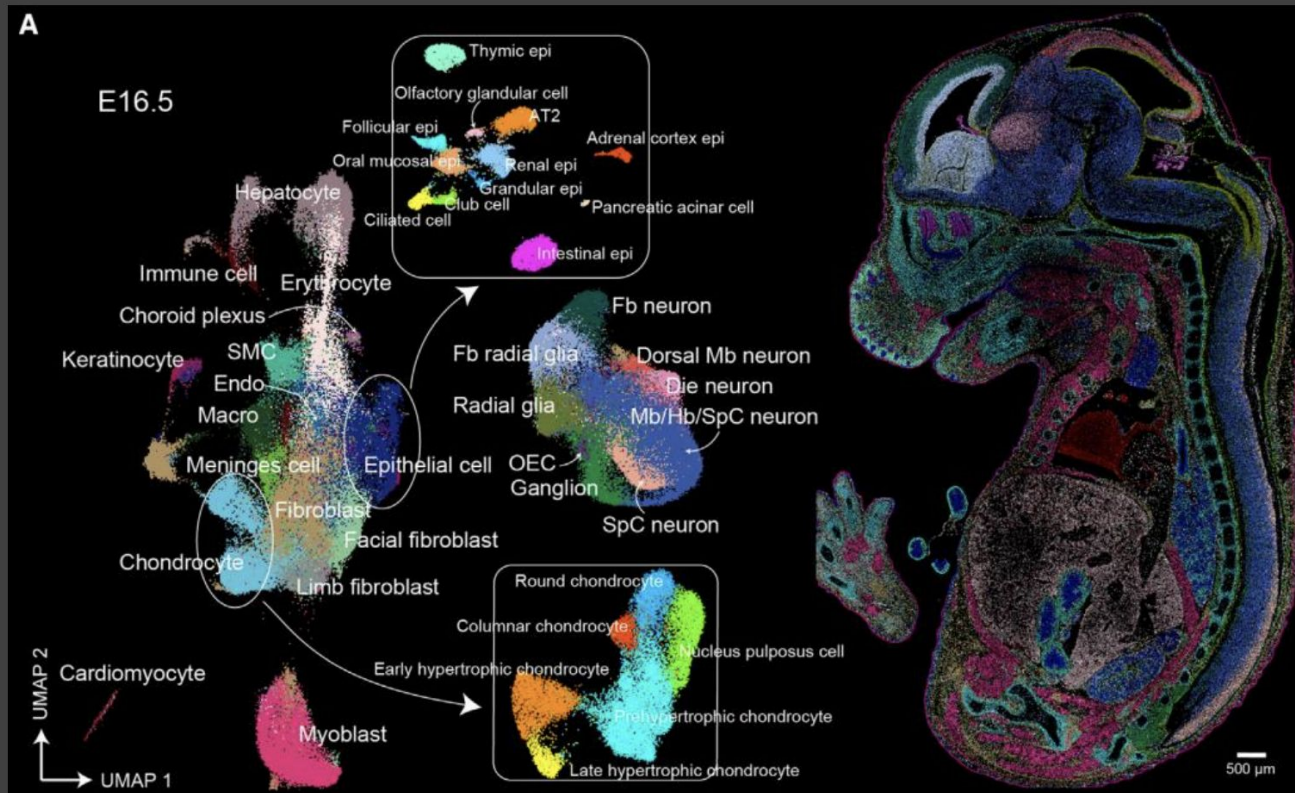
Stereo-seq



220nm DNA nanoballs

Chen et al. Cell 2022

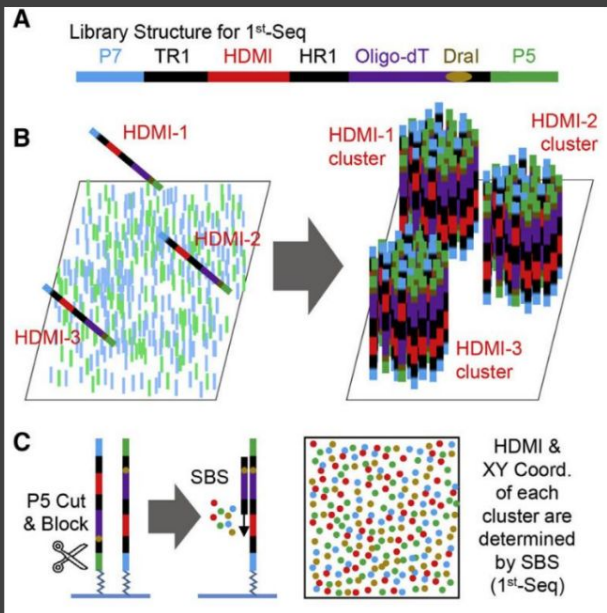
Stereo-seq



Chen *et al.* Cell 2022

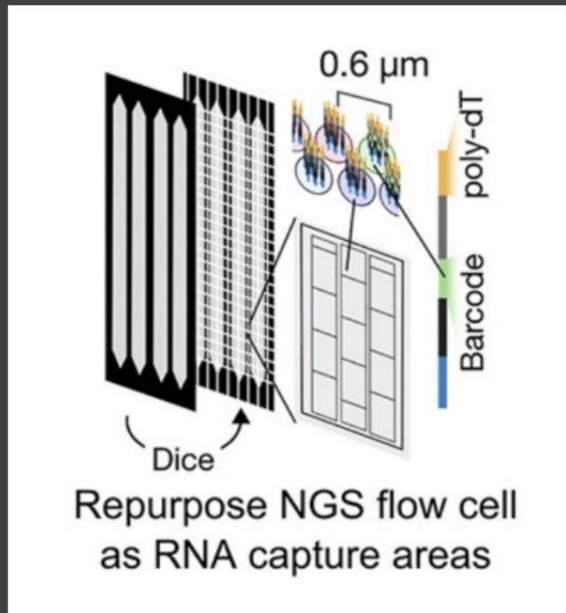
Seq-Scope, Open-ST, Nova-ST

Seq-Scope



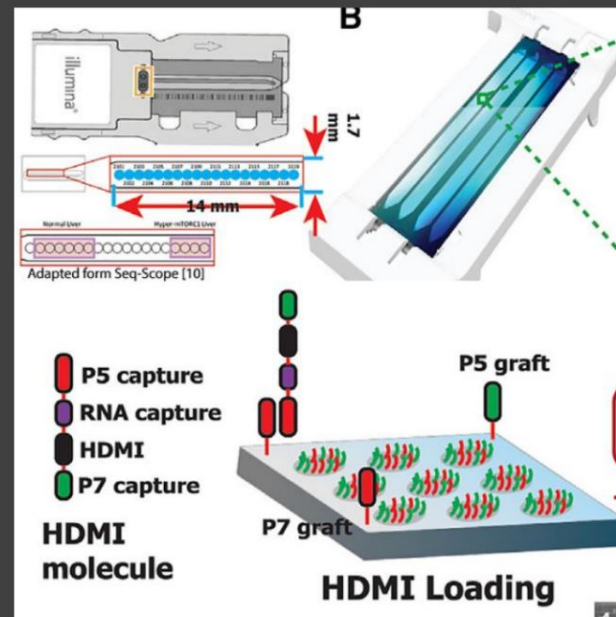
Cho *et al.* 2021 Cell

Open-ST



Schott *et al.* 2024 Cell Reports Methods

Nova-ST



Poovathingal *et al.* 2024 Cell Reports Methods

Spatial Sequencing



Methods: ST, Slide-seq, Stereo-seq.
Commercial: 10X Visium, Curio Seeker



Resolution: Spot size 220nm - 100um (but RNA diffuses)



Detection efficiency: 0.1 - 5%



Gene throughput: Full transcriptome

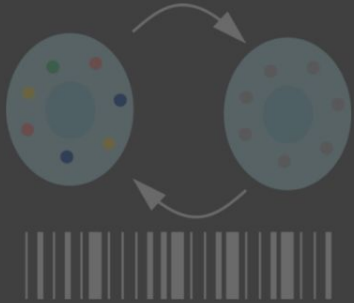


Spatial throughput: several mm² – several cm²

spatial RNA detection

Microscopy

Barcoded
FISH



in situ Sequencing



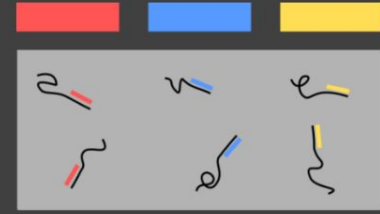
Sequencing

Spatial Sequencing



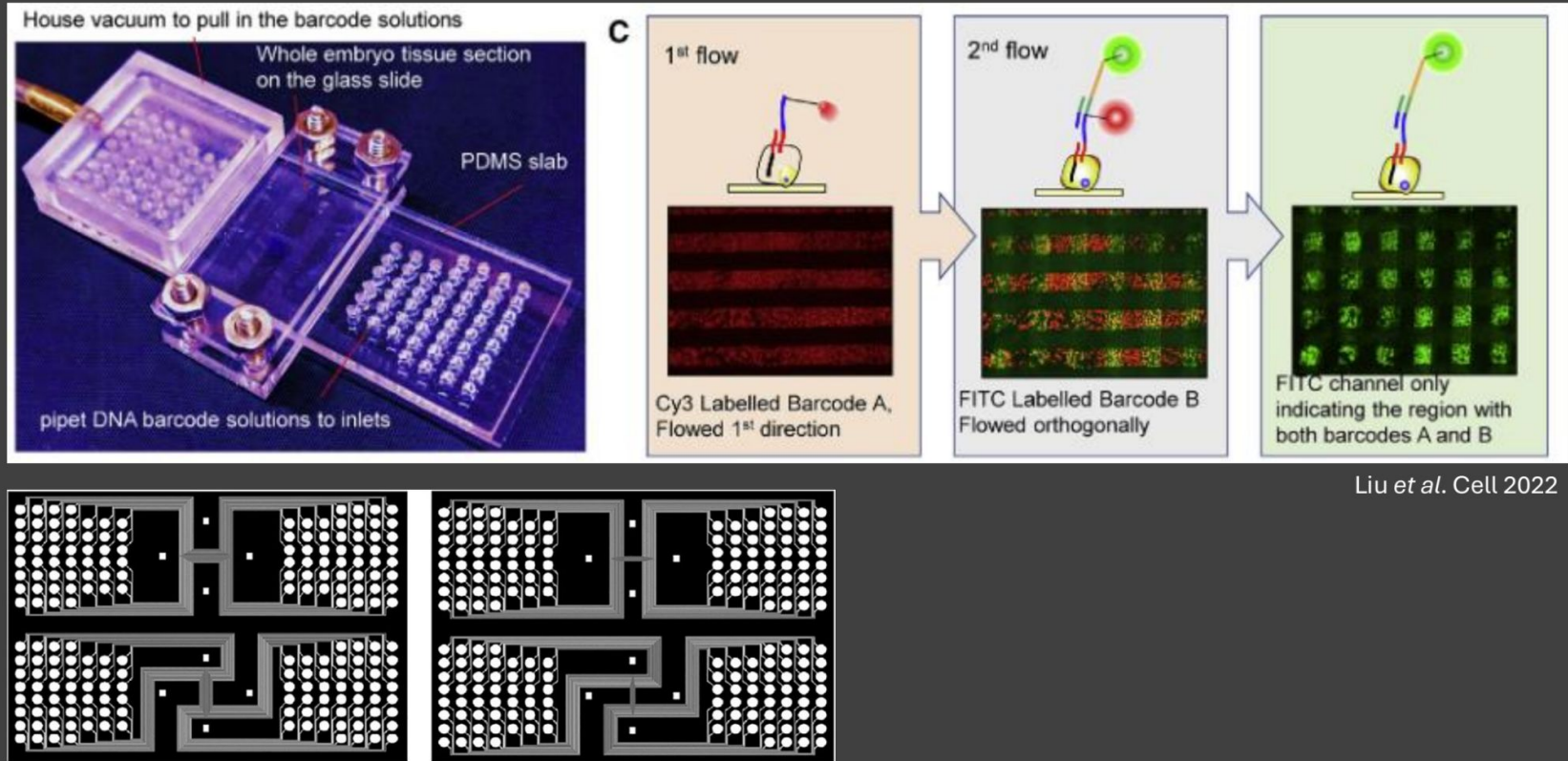
RNA moves

Spatial tagging



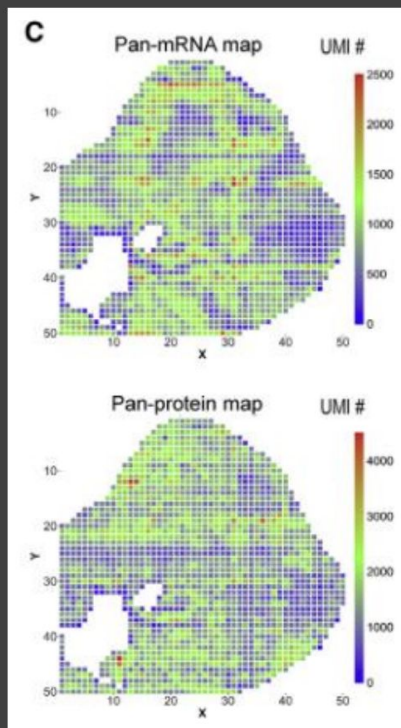
Barcodes move

DBiT-seq

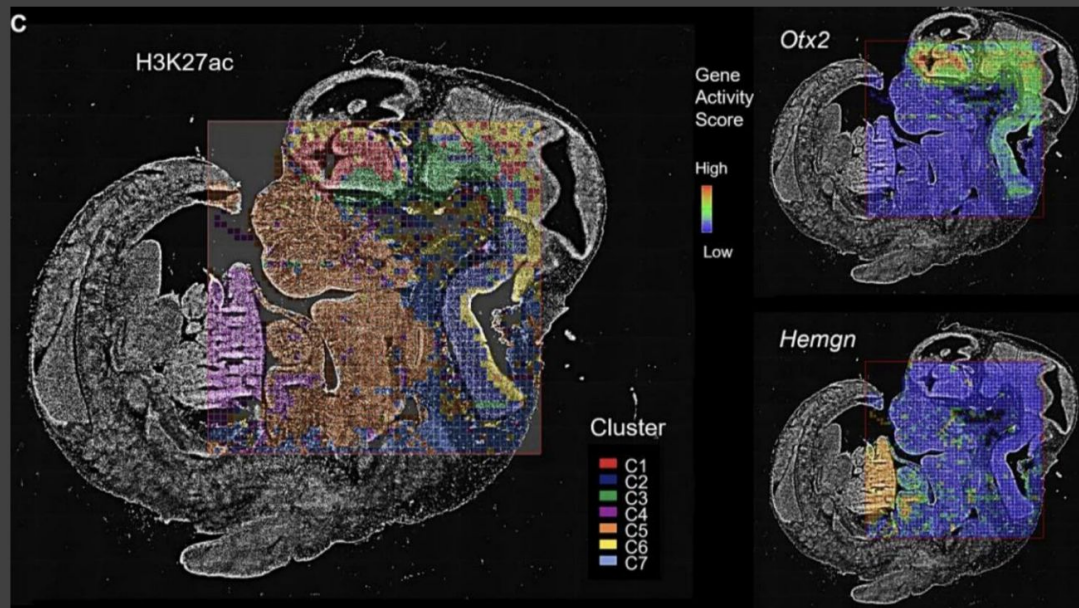


Liu *et al.* Cell 2022

DBiT-seq

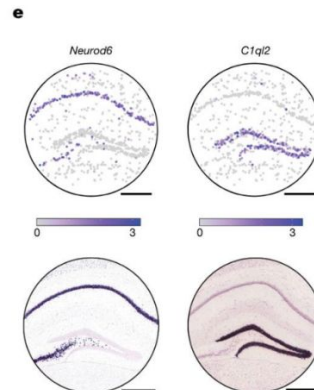
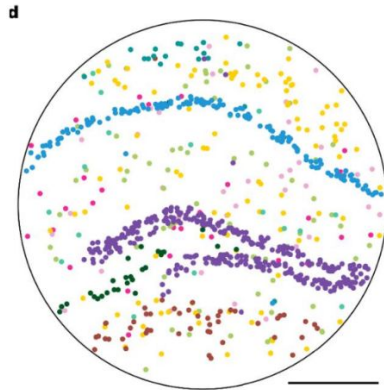
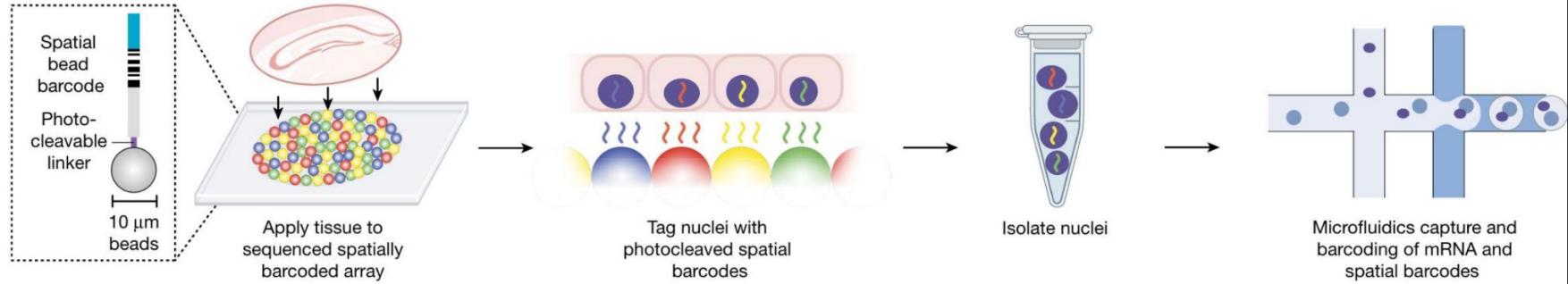


Liu et al. Cell 2022



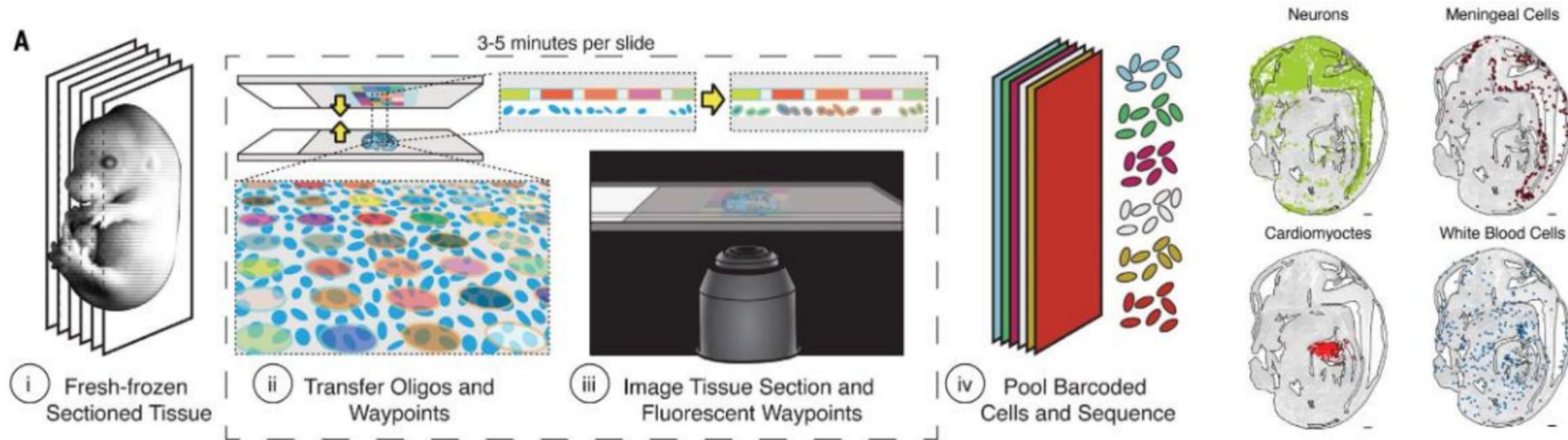
Deng et al. Science 2022

Slide-tags



Russel et al. Nature 2024

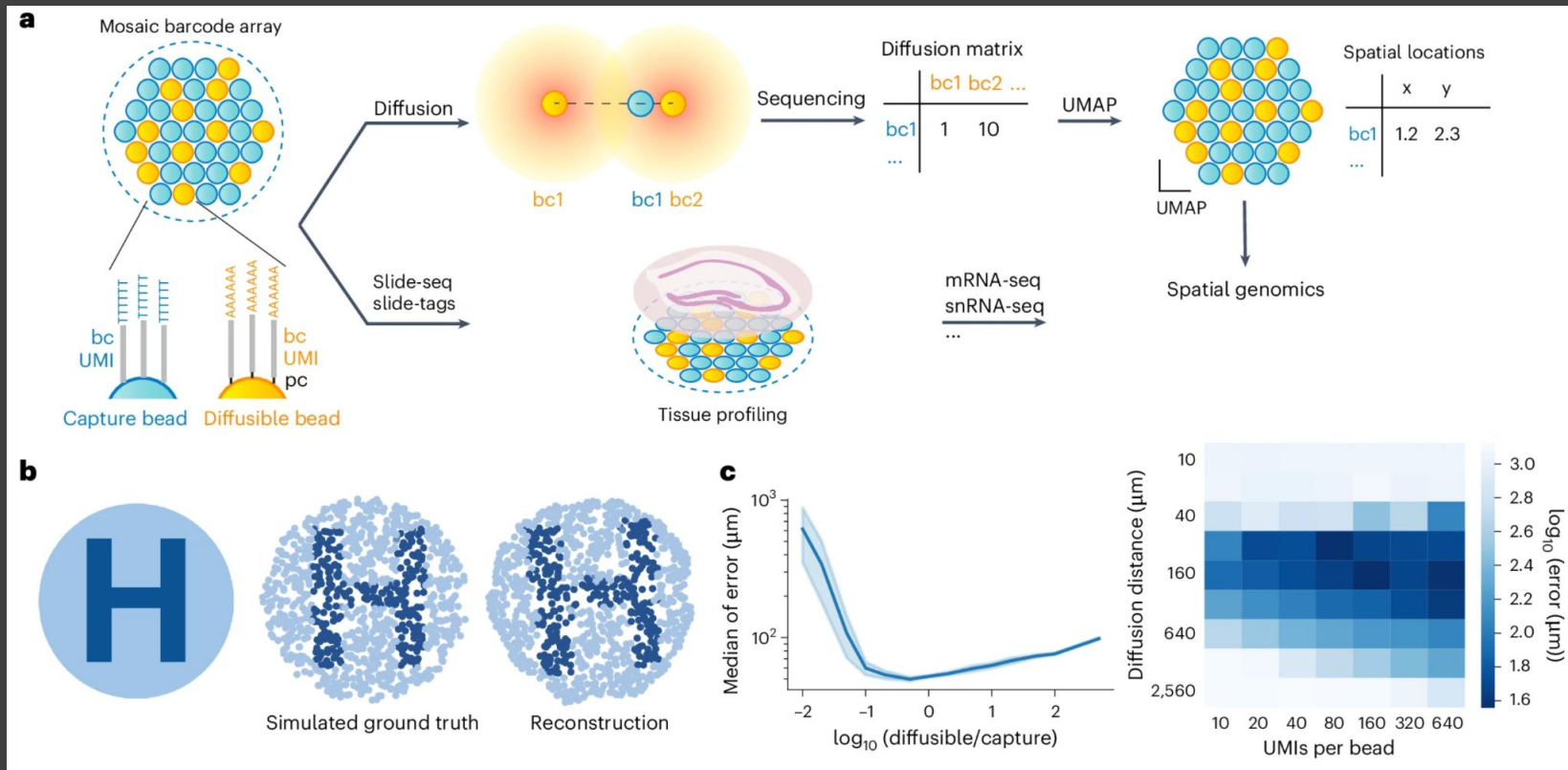
sci-Space



73um spots, 222um between spots, 2.2% of nuclei sampled

Srivatsan *et al.* Science 2022

Imaging-free spatial transcriptomics



Spatial Tagging



Methods: DBiT-seq, Slide-tags, sci-Space
Commercial: DBiT-seq, Curio Trecker



Resolution: 10 – 100 μ m



Detection efficiency: 1 - 30%



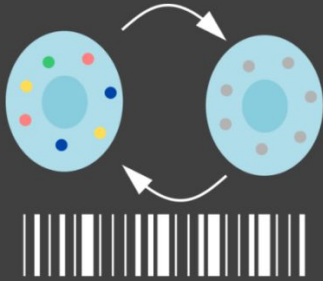
Gene throughput: Full transcriptome



Spatial throughput: several mm² – cm²

Spatial RNA detection

Barcoded FISH



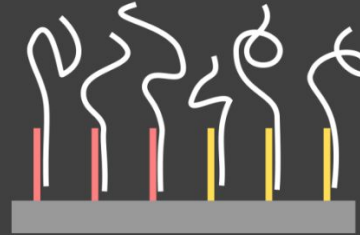
- MERFISH
- Vizgen - MERFISH
- seqFISH
- Spatial Genomics - GenePS
- EEL-FISH
- HybISS
- 10X - Xenium
- Nanostring/Bruker - CosMx
- Resolve - Mol. Cartography

in situ Sequencing



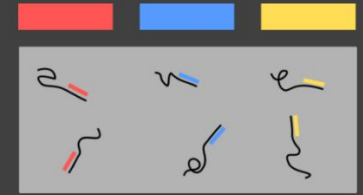
- ISS
- STARmap
- StellarOmics
- Singular genomics - G4X

Spatial Sequencing



- Spatial Transcriptomics
- 10X - Visium (HD)
- Slide-seq
- Curio - Seeker
- Stereo-seq
- BGI STOmics - Stereo-seq
- Seq-Scope, Open-ST, Nova-ST

Spatial tagging



- DBiT-seq
- AtlasXomics
- Slide-tags
- Curio - Trekker

Spatial Transcriptomics



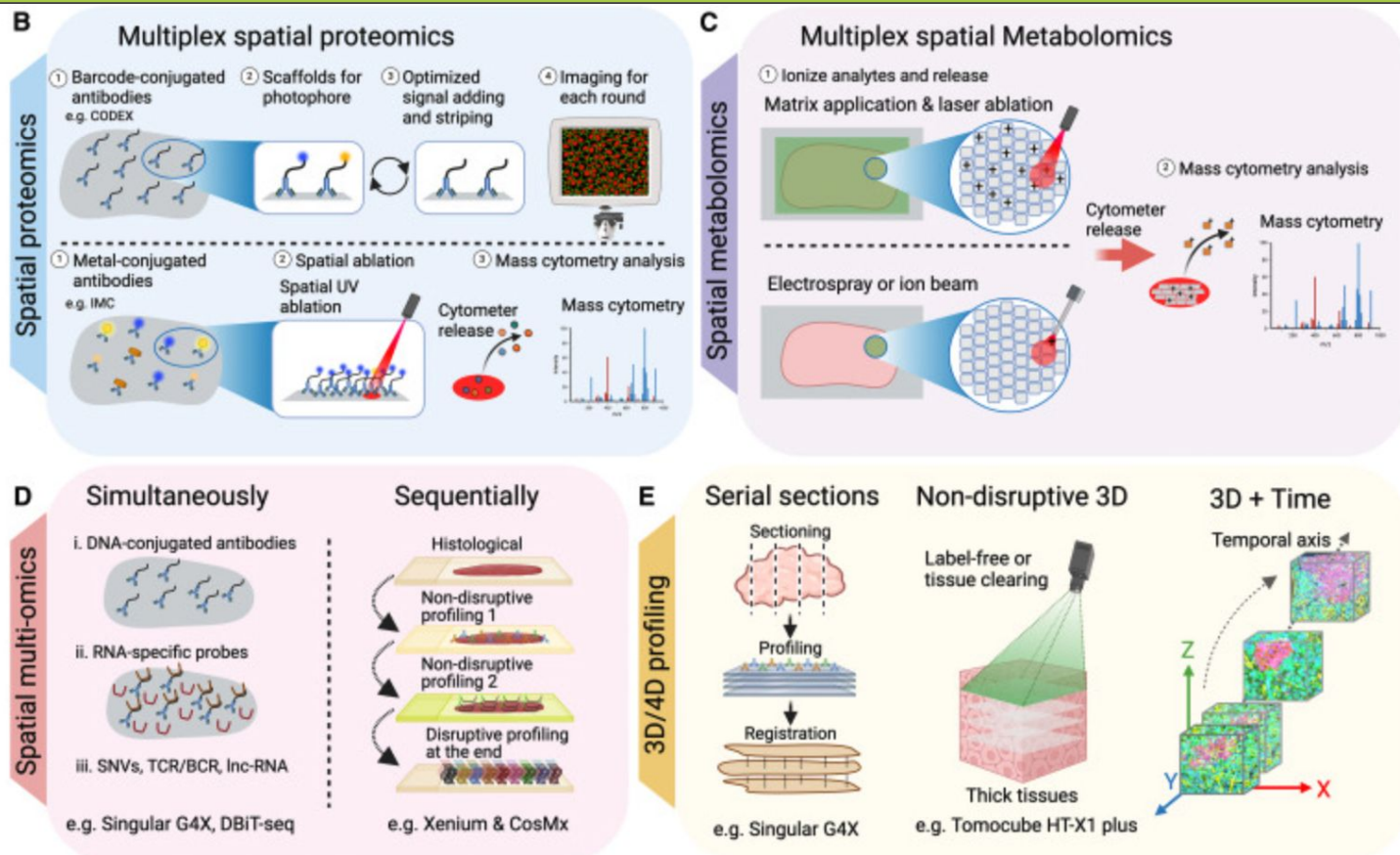
	Barc. FISH	Amp. FISH	ISS	Sequencing	Tagging
Resolution	Diffraction 150-300nm	Amplicon size 0.5-1um	Amplicon size 0.5-1um	2-10 um	10-100 um
Gene detection	70-95%	10-50%	10-50%	5-15%	10-30%
Gene throughput	100-19,000	100-10,000	100-10,000	Whole transcriptome	Whole transcriptome
Area covered	mm ²	>cm ²	mm ² to cm ²	cm ²	cm ²

Price? Time?

Throughput vs detection in SRT

- In general, higher number of genes gives lower detection per gene
- For quantification of celltypes:
 - A celltype specific gene panel is better
 - E.g. amount of immune infiltration in tumors.
- For discovery of novel markers:
 - Need full transcriptome
 - Or combination of scRNAseq/bulkRNAseq and SRT with few markers

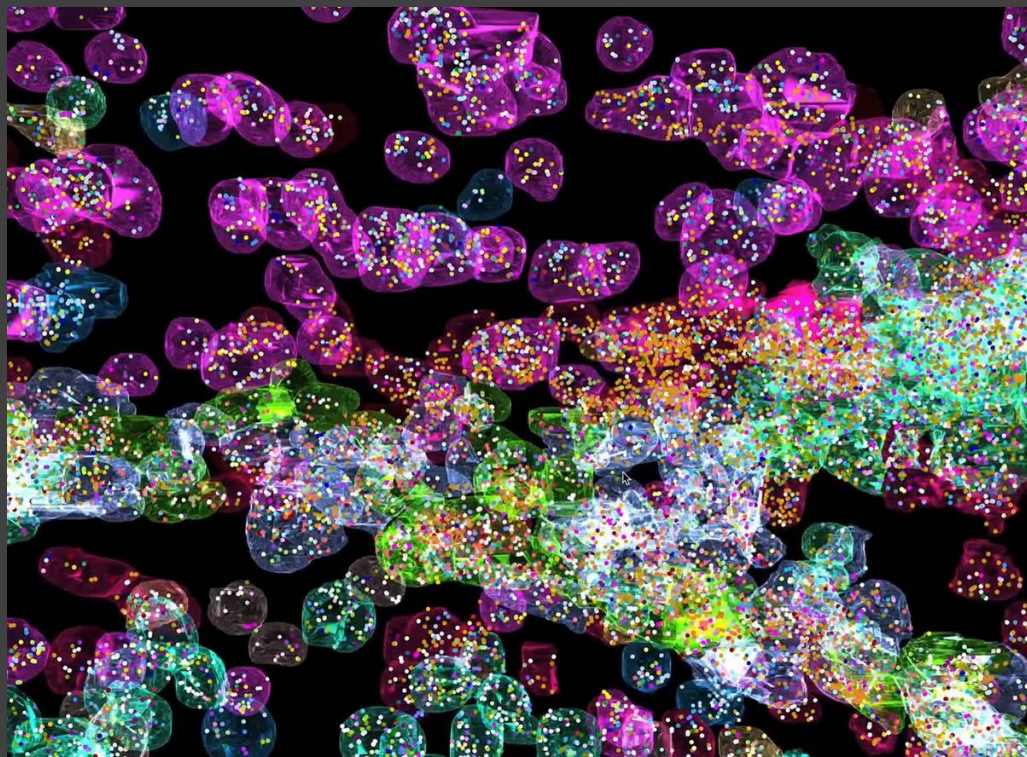
Spatial multi-omics



3D spatial omics

Stellaromics' Pyxa platform can profile over 1,000 genes in a 20µm-100µm thick section.

Many labs working on tissue clearing and method optimization for both ISS and FISH based 3D-imaging.



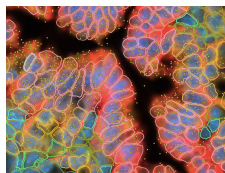
stellaromics.com

Spatial Omics Platform at SciLifeLab



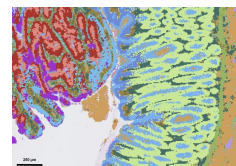
In Situ Sequencing Unit Xenium

Imaging based **transcriptomics**

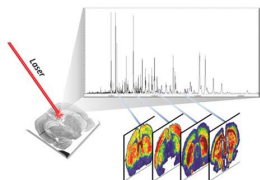


Genomics Unit VisiumHD

Sequencing based **transcriptomics**

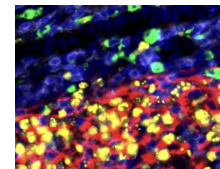


Spatial Mass Spec Unit
MSI based **proteomics**, **lipidomics** and **metabolomics**.



Spatial Proteomics Unit Akoya / CODEX

Immunofluorescence based **proteomics**



Capabilities at ScilifeLab Spatial platform

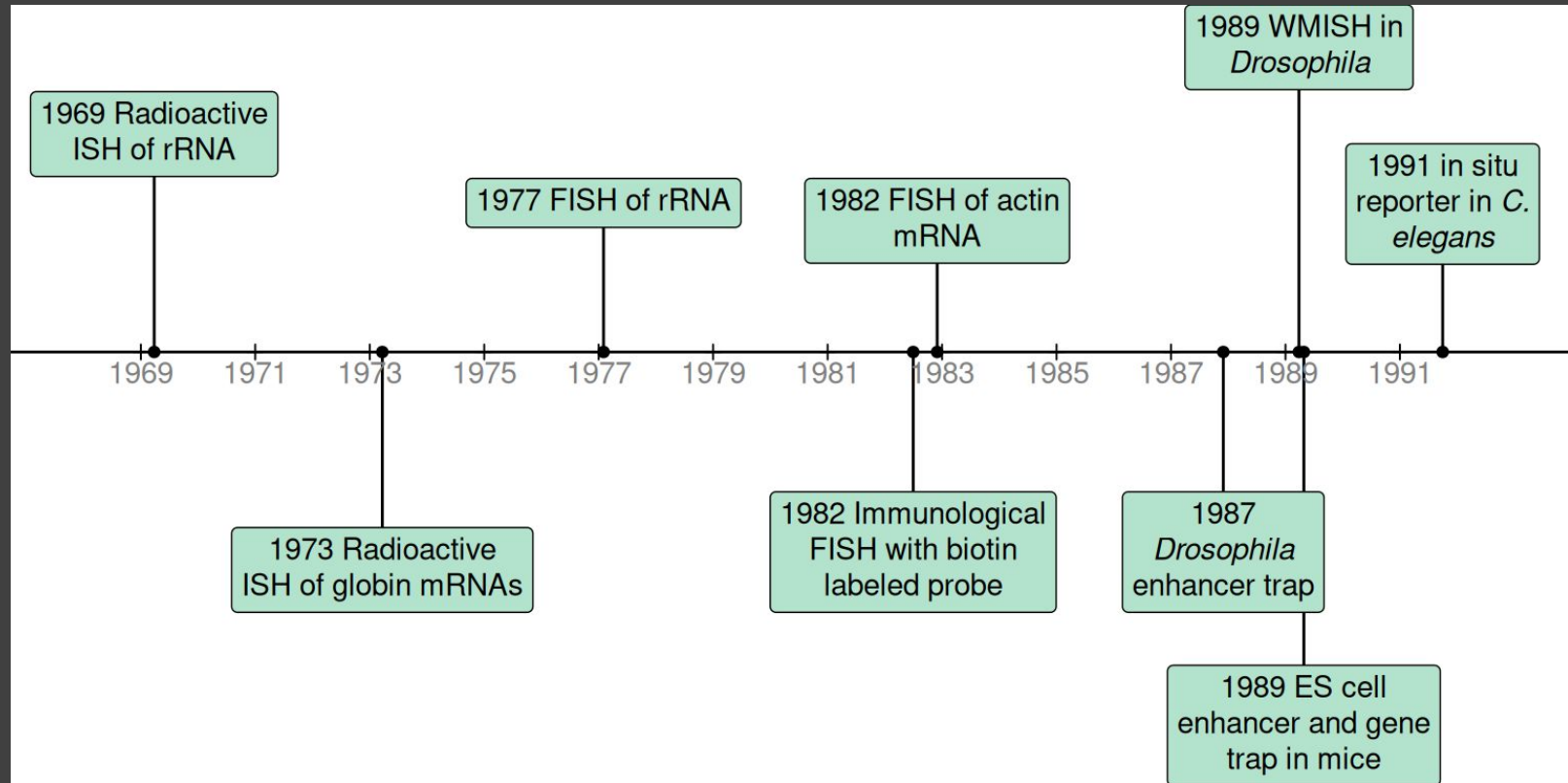
Unit and technology	Omics type	Readout	Complexity	Sample requirements		Spatial resolution
				Fresh-frozen	FFPE	
ST - 10X Visium	RNA	NGS Sequencing Microscopy of tissue section	Transcriptome wide	✓	✓	Anatomical feature of 55 um
ISS - 10X Xenium	RNA	Microscopy	Predesigned panels up to 500 genes Custom panels	✓	✓	Sub-micron
ISS - In situ sequencing (in house)	RNA	Microscopy	Custom panels of up to 400 genes	✓	✓	Sub-micron
SP - sm RNA/DNA FISH	RNA/DNA	Microscopy	Custom in house probe design 1-6 targets	✓	✓	Sub-micron
SMS - MALDI Imaging	Small molecules < 20 kDa	Mass spectrometry	Multiplexed, targeted or untargeted	✓	✗	Anatomical feature of 5 um
SMS - DESI Imaging	Small molecules < 20 kDa	Mass spectrometry	Multiplexed, targeted or untargeted	✓	✗	Anatomical feature of 20 um
SP - Akoya Phenocycler	Proteins	Microscopy	Pre-conjugated antibodies > 40 proteins Custom conjugations	✓	✓	Sub-micron
SP - Akoya Phenocycle/Opal	Proteins	Microscopy	Panels of 6 antibodies (TSA amplification)	✓	✓	Sub-micron
SP - Lunaphore COMET	Proteins	Microscopy	Custom antibody panels of 6-30 proteins	✓	✓	Sub-micron

Disclaimer:

New methods come out constantly and all methods are not mentioned in this presentation. We try to cover the most relevant ones but may miss some protocols in the presentations.

Questions?

Spatially resolved transcriptomics (SRT)



Spatially resolved transcriptomics (SRT)

